A Mechanistic Study of Scale-Dependent Optimization Stagnation in C^1 Piecewise-Quadratic Rectifiers Under Class Imbalance: Bias-Offset Equivalence, Layer-Wise Gradient Dynamics, and Scale-Adaptive Remedies

Preecha Noiumkar[1,*] and **Thawatchai Chomsiri**[1]

[1]Research Center of Information Technology for the Future, Department of Information Technology, Faculty of Informatics, Mahasarakham University, Mahasarakham 44150, Thailand

[*]Corresponding author. Emails: `preecha.n@msu.ac.th` (P. Noiumkar), `thawatchai.c@msu.ac.th` (T. Chomsiri)

Abstract

Piecewise-quadratic C^1-continuous rectifiers such as Smooth ReLU (SmeLU, centered symmetrically on $[-\beta, +\beta]$) replace ReLU's non-differentiable kink with a parabolic transition. Recently proposed Smooth RectLU and its trainable variant AS-RectLU anchor this transition asymmetrically on $[L, L+W]$ with default $L = 0, W = 2$. Algebraically, Smooth RectLU satisfies the exact identity $f_S(x; L, W) = \mathrm{SmeLU}_{\beta=W/2}(x - L - W/2)$, meaning that initializing at $L = 0, W = 2$ imposes an implicit negative bias offset of $\Delta b = -1$ and a zero origin derivative ($f'(0^+) = 0$) relative to symmetric SmeLU. In this mechanistic study, we derive second-moment propagation equations distinguishing pre-activation variance $q_\ell = \mathrm{Var}(x_\ell)$ from post-activation variance $v_\ell = \mathrm{Var}(a_\ell)$ and measure layer-wise weight gradient norms across $N = 10$ random seeds to explain why 2D convolutional networks using asymmetric quadratic rectifiers stagnate under severe class imbalance ($10:1$ to $200:1$) when pre-activation scales are small relative to transition width ($\sigma_\ell / W \ll 1$). At initialization on low-amplitude 2D inputs ($\sigma_x = 0.22$), Smooth RectLU ($L = 0, W = 2$) contracts fourth-layer pre-activation variance q_4 to 1.99×10^{-10} (post-activation $v_4 = 4.67 \times 10^{-21}$) and first-layer weight gradient norms to $7.17 \times 10^{-11} \pm 4.87 \times 10^{-11}$. Epoch-by-epoch tracking reveals a variance-escape bifurcation: seeds whose dense pre-activation variance expands past the quadratic ramp ($q_4 \in [3.54, 3.90]$ by Epoch 30) achieve full convergence (MCC(0.50) $\in [0.83, 0.89]$), whereas seeds remaining below unity ($q_4 \in [0.46, 0.75]$) remain trapped at MCC(0.50) $= 0.0000$. This vulnerability generalizes across Adam, RMSprop (MCC $= 0.1570$ vs. 0.5762 for σ-AS-RectLU), and SGD with momentum (AUROC $= 0.5000$ vs. 0.8159). Across 2×2 factorial experiments (Offset $L \in \{0, -1\} \times$ Scale σ_x, $N = 10$ paired seeds, sample SD), we observe significant two-way interactions on synthetic 2D motifs (DiD AUROC $= +0.3633 \pm 0.1248, p = 0.0020$) and on both variant-scrubbed (DiD AUROC $= +0.4029 \pm 0.0331, p = 0.0020$) and unscrubbed natural (DiD AUROC $= +0.0416 \pm 0.0384, p = 0.0137$) $10:1$ imbalanced 2D NIST GIAB HG001 BAM read-pileup benchmarks, where Symmetric σ-AS-RectLU (MCC(0.50) $= 0.9366$ and 0.9353, and 0.9797 ± 0.0328 across $N = 30$ paired seeds and $K = 6$ disjoint genomic test sub-blocks with 30/30 seed wins) matches homogeneous ReLU (0.9828 ± 0.0327) and modular 3-step RMSNorm+SmeLU+Rescale while reducing measured 2D pileup training latency by 11.9% (33.6% faster than LayerNorm+SmeLU) and eliminating low-scale stagnation.

Keywords: Activation function; Piecewise-quadratic rectifier; SmeLU; Smooth RectLU; Gradient propagation; Class imbalance; Matthews correlation coefficient

1 Introduction

Convolutional neural networks (CNNs) deployed for rare-event screening—motivated by applications such as genomic variant screening in high-throughput sequencing [1, 2, 3, 6, 35] and visual or tabular anomaly detection [13, 14]—must learn minority representations under severe class imbalance ($10:1$ to $200:1$). In such regimes, overall accuracy is dominated by majority negatives, requiring evaluation via the Matthews correlation coefficient (MCC) [8, 9], area under the precision-recall curve (AUPRC) [10, 11], area under the ROC curve (AUROC), observed per-seed empirical ranges, and explicit confusion matrix counts [12, 14]. Because severe prior imbalance shifts raw output logits negatively ($\ln(p_+/p_-) \approx -2.3$ to -5.3), evaluation must distinguish between *threshold miscalibration* (where ranking AUROC is high and calibrating τ^* on a validation set restores MCC) and *optimization stagnation / seed trapping*

(where multi-layer variance and gradient contraction stalls parameter updates, leaving AUROC near chance or trapping a subset of random seeds in sub-optimal states).

Piecewise-quadratic C^1 rectifiers replace the non-differentiable kink of ReLU [22, 23] with a parabolic transition using elementary arithmetic operations. Shamir et al. [41] introduced Smooth ReLU (SmeLU) by centering a quadratic transition symmetrically over $[-\beta, +\beta]$ (threshold $L = -\beta$, width $W = 2\beta$). More recently, Chomsiri and Noiumkar [15] proposed RectLU and its C^1 variant Smooth RectLU, anchoring the transition over $[L, L+W]$ with an asymmetric default of $L = 0, W = 2$. Trainable activation schemes [26, 30, 39] attempt to learn shape parameters during training.

A fundamental algebraic identity governs this entire family: for any threshold $L \in \mathbb{R}$ and transition width $W > 0$, Smooth RectLU is an exact shifted copy of symmetric SmeLU:

$$f_S(x; L, W) = \mathrm{SmeLU}_{\beta=W/2}\left(x - L - \frac{W}{2}\right). \tag{1}$$

In any layer with a learnable bias vector $\mathbf{b}$, applying $f_S(\mathbf{Wh} + \mathbf{b}; L, W)$ is algebraically identical to applying symmetric $\mathrm{SmeLU}_{\beta=W/2}(\mathbf{Wh} + \mathbf{b}_{\text{eff}})$ with shifted bias $\mathbf{b}_{\text{eff}} = \mathbf{b} - (L + W/2)\mathbf{1}$. Despite having identical representational capacity, 2D convolutional networks initialized with standard zero biases ($\mathbf{b}_0 = \mathbf{0}$), $L = 0$, and wide W relative to input scale σ_x suffer from a **synergistic interaction (H1 × H2)** when trained on low-amplitude, imbalanced data:

- **Factor H1 (Scale-Width Mismatch $\sigma_\ell \ll W$ and Loss of Positive Homogeneity):** Unlike ReLU ($f(\alpha x) = \alpha f(x)$), any fixed-width piecewise-quadratic rectifier breaks scale invariance. When pre-activations have small amplitude relative to transition width ($\sigma_\ell / W \ll 1$), signals fall inside the quadratic ramp where forward variance contracts super-linearly ($q_{\ell+1} \propto (q_\ell / W^2) q_\ell$) [18, 19, 20].
- **Factor H2 (Implicit Negative Bias Offset $\Delta b = -(L + W/2)$ Shifting Zero-Mean Pre-Activations to the Zero-Derivative Boundary $f'(0^+) = 0$):** Because $f_S(x; L, W) = \mathrm{SmeLU}_{W/2}(x + \Delta b)$, setting $L = 0$ ($\Delta b = -W/2$) shifts zero-mean pre-activations ($x \approx 0$) from the center of SmeLU's ramp ($f'(0) = 0.5$) directly onto the left dead boundary ($f'(0^+) = 0$). Initializing all layer biases to $b_0 = +W/2 = +1.0$ while keeping $L = 0, W = 2$ restores Epoch 0 layer-wise variances and weight gradients to the exact values of symmetric SmeLU ($L = -1, W = 2, b_0 = 0$). However, because learnable biases $\mathbf{b}$ are updated by majority-dominated gradients during training whereas a fixed threshold $L = -W/2$ is not, fixed negative centering ($L = -W/2$) is more stable over 20 epochs (0.7406 ± 0.0687) than bias initialization alone (0.6014 ± 0.3248). Crucially, neither H1 nor H2 alone induces 20-epoch 2D failure: SmeLU ($\Delta b = 0$) converges at low scale ($\sigma_x = 0.22$), and Smooth RectLU ($L = 0, \Delta b = -1$) converges at unit scale ($\sigma_x = 1.00$).

This paper is positioned as a **mechanistic investigation of scale- and offset-dependent optimization failure** in smooth piecewise-quadratic rectifiers, rather than a claim that a new activation universally outperforms ReLU. Because standard ReLU is positively homogeneous ($f(\alpha x) = \alpha f(x)$), it is naturally immune to scale-width mismatch. Our goal is to characterize why fixed-width C^1 quadratic rectifiers fail when initialized asymmetrically at $L = 0$ under low signal scales, and to evaluate how negative threshold centering ($L < 0$), explicit normalization (`LN-no-affine + SmeLU`), or an embedded scale-adaptive formulation (**Symmetric Per-Sample σ-AS-RectLU**, equivalent to parameter-free RMSNorm + SmeLU + inverse rescaling) restore convergence to parity with ReLU.

The primary contributions of this work are:

1. **Theoretical Equivalence, Pre/Post-Activation Second-Moment Recurrence, and Multi-Seed Layer-Wise Gradient & Variance-Bifurcation Dissection:** We establish the bias-shift identity $f_S(x; L, W) = \mathrm{SmeLU}_{W/2}(x - L - W/2)$, distinguish pre-activation variance $q_\ell = \mathrm{Var}(x_\ell)$ from post-activation variance $v_\ell = \mathrm{Var}(a_\ell)$ across layers, and measure multi-seed ($N = 10$) layer-wise variances, zero-gradient fractions, and weight gradient norms ($\|\nabla_{\mathbf{W}_\ell} \mathcal{L}\|_2$) across convolutional layers (Table 5, Fig. 2). Fine-grained epoch-by-epoch tracking (Epochs 0–30) demonstrates that seed escape is associated with and causally supported by a fourth-layer pre-activation variance bifurcation ($q_4 > 3.5$ for escaped seeds vs. $q_4 < 0.8$ for trapped seeds).
2. **Controlled 2×2 Factorial Dissection ($N = 10$ Seeds), Cross-Optimizer Generalization (Adam, RMSprop, SGD), and 50/100-Epoch Seed-Trapping Analysis:** Across 1D and 2D sparse motif generators ($N = 10$ seeds, sample SD, Tables 2–9), we demonstrate via difference-in-differences on ranking AUROC (DiD AUROC $= +0.3633, p = 0.0020$) that 2D stagnation requires

the co-occurrence of $L = 0$ and $\sigma_x \ll W$, persists across all 10 seeds at $\epsilon = 10^{-12}$, and generalizes to RMSprop and momentum SGD.

3. **Multi-Domain Evaluation and Dual (Scrubbed & Unscrubbed Natural) Real NIST GIAB HG001 2×2 Optimization Benchmarks ($N = 10$ Seeds):** Beyond step-imbalanced MNIST ($100:1$) and Tabular ($200:1$, Table 8), we evaluate real NIST GIAB HG001 sequencing data under strict controls (Table 11, Fig. 5): (a) a 1D k-mer coverage-shortcut audit showing that per-sample unit-L2 normalization removes read-depth norm shortcuts, and (b) $10:1$ imbalanced 2D BAM read-pileup optimization benchmarks ($N = 10$ seeds, disjoint genomic blocks) evaluated on both variant-scrubbed reference windows (DiD AUROC $= +0.4029 \pm 0.0331, p = 0.0020$) and **unscrubbed natural reference windows** (DiD AUROC $= +0.0416 \pm 0.0384, p = 0.0137$, where window-wide and center-locus secondary-allele heuristics perform at chance: AUROC $= 0.5075$ and 0.4882).

2 Related Work

2.1 Piecewise-Quadratic, Smooth, and Trainable Rectifiers

ReLU [22, 23] resolved vanishing positive gradients at the cost of the "dying ReLU" phenomenon [24]. Leaky ReLU [25], PReLU [26], ELU [17], SELU [27], GELU [16], SiLU [28], and Mish [29] modify the negative or near-zero regime; Dubey et al. [30] provide a comprehensive survey. To achieve C^1 continuity without transcendental functions, Shamir et al. [41] introduced symmetric SmeLU on $[-\beta, +\beta]$, while Chomsiri and Noiumkar [15] proposed RectLU and C^1 Smooth RectLU on $[L, L+W]$ with default $L = 0, W = 2$. Trainable activation frameworks [26, 39] attempt to learn shape parameters via backpropagation, but suffer high variance or parameter freeze when initialized inside flat derivative regions.

2.2 Signal Propagation, Normalization, and Imbalanced Classification

Glorot and Bengio [31] and He et al. [26] derived variance-preserving weight initializations for linear and ReLU layers, while Mishkin and Matas [32] (LSUV) and Scabini et al. [40] established data-driven initialization schemes. Mean-field theory [18, 19, 20] characterizes how pre-activation variance evolves with depth. Batch Normalization (BN) [33], Layer Normalization (LN), and RMS Normalization enforce standardized pre-activation scales externally. Under severe class imbalance [13, 14], cost-sensitive and focal objectives [21, 42] re-weight minority errors against gradient starvation [34]. Rigorous evaluation requires MCC [8, 9], AUPRC [10, 11], confusion counts [12], and exact non-parametric statistical tests with Holm–Bonferroni correction [36].

3 Theoretical Analysis and Scale-Adaptive Formulation

3.1 Exact Algebraic Equivalence to Bias-Shifted SmeLU and RectLU Continuity

For threshold $L \in \mathbb{R}$ and transition width $W > 0$, two variants of non-C^1 piecewise-quadratic RectLU exist depending on the linear branch for $x \geq L + W$ [15]:

$$
\begin{aligned}
f_{R,\text{shift}}(x; L, W) &= \begin{cases} 0, & x < L \\ x(x-L)/W, & L \leq x < L+W \\ x - L - W/2, & x \geq L+W \quad (D^0 \text{ value jump at } x = L+W), \end{cases} \\
f_{R,\text{cont}}(x; L, W) &= \begin{cases} 0, & x < L \\ x(x-L)/W, & L \leq x < L+W \\ x, & x \geq L+W \quad (C^0 \text{ continuous at } L = 0). \end{cases}
\end{aligned} \tag{2}
$$

In both forms, the derivative inside $[L, L+W]$ is $f'_R(x) = (2x-L)/W$. At $x = 0$, setting $L = 0$ yields a vanishing right-hand derivative ($f'_R(0^+) = 0$), whereas shifting $L < 0$ provides a strictly positive origin slope $f'_R(0) = -L/W > 0$. Exact closed-form Gaussian integration over $x \sim \mathcal{N}(0,1)$ confirms that at $L = 0, W = 2$, the shifted D^0 form $f_{R,\text{shift}}$ has forward second-moment gain $\mathbb{E}[f^2]/\mathbb{E}[x^2] = 0.2145$, whereas the unshifted continuous C^0 form $f_{R,\text{cont}}$ has gain 0.2998 (Table 1). Smooth RectLU (C^1)

replaces the middle segment with a tangent parabola:

$$f_S(x; L, W) = \begin{cases} 0, & x < L \\ (x-L)^2/(2W), & L \le x < L+W \\ x - L - W/2, & x \ge L+W. \end{cases} \tag{3}$$

Comparing Eq. (3) with Shamir et al.'s symmetric SmeLU [41] of half-width $\beta = W/2 > 0$, defined as $\mathrm{SmeLU}_\beta(z) = (z+\beta)^2/(4\beta)$ on $[-\beta, +\beta]$, 0 for $z < -\beta$, and z for $z \ge \beta$, substituting $z = x - L - W/2$ yields Eq. (1) identically. Thus, $f_S(x; L, W)$ is governed by its **effective bias offset** $\Delta b = -(L+W/2)$ and its **transition width** $W = 2\beta$ (Fig. 1). Equation (1) implies that at initialization (Epoch 0), shifting L from 0 to $-W/2$ is algebraically identical to adding a constant bias shift $b_0 = +W/2$ to every layer; we use this identity as an empirical sanity check in Section 5.1 (Table 5) while also testing how learnable biases diverge from fixed $L = -W/2$ during training (Table 8).

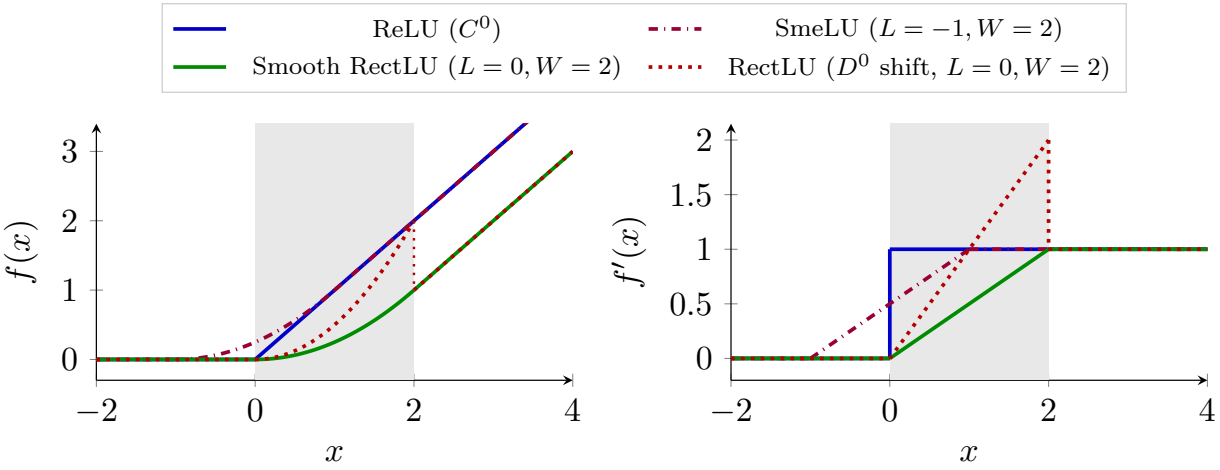


Figure 1: Activation functions (left) and derivatives (right) comparing ReLU (C^0), symmetric SmeLU ($C^1, L = -1, W = 2, \Delta b = 0$), Smooth RectLU ($C^1, L = 0, W = 2, \Delta b = -1$), and shifted RectLU ($D^0$, whose derivative rises as $2x/W = x$ to 2.0 on $[0, 2)$ before dropping to 1.0 at $x = 2$). The shaded band marks $[0, 2]$ where $f'_S(x) = x/W$ rises linearly from zero.

3.2 Pre- and Post-Activation Second-Moment Propagation Under Scale-Width Mismatch ($\sigma_\ell \ll W$)

To maintain strict correspondence between theoretical signal propagation and empirical layer measurements (Table 5), we explicitly distinguish the **pre-activation second moment** $q_\ell = \mathbb{E}[x_\ell^2] \approx \mathrm{Var}(x_\ell)$ from the **post-activation second moment** $m_\ell = \mathbb{E}[a_\ell^2]$ and **post-activation variance** $v_\ell = \mathrm{Var}(a_\ell)$ where $a_\ell = f_S(x_\ell; L, W)$. For $x_\ell \sim \mathcal{N}(0, q_\ell)$ at $L = 0$ in the small-signal regime ($\sqrt{q_\ell} \ll W$), $f_S(x_\ell) = x_\ell^2/(2W)$ for $x_\ell > 0$ and 0 for $x_\ell \le 0$. Because $\mathbb{E}[x_\ell^4 \mathbb{I}(x_\ell > 0)] = \frac{3}{2} q_\ell^2$ and $\mathbb{E}[x_\ell^2 \mathbb{I}(x_\ell > 0)] = \frac{1}{2} q_\ell$, the post-activation second moment m_ℓ, post-activation variance v_ℓ, and backward derivative second moment are:

$$m_\ell = \mathbb{E}[a_\ell^2] = \frac{3\, q_\ell^2}{8W^2}, \qquad v_\ell = \mathrm{Var}(a_\ell) = \frac{5\, q_\ell^2}{16W^2}, \qquad \mathbb{E}[f'_S(x_\ell)^2] = \frac{q_\ell}{2W^2} \qquad (L = 0, \sqrt{q_\ell} \ll W). \tag{4}$$

Under He initialization ($\mathrm{fan_in} \cdot \sigma_w^2 = 2$) [26] with zero initial biases, pre-activation variance in layer $\ell{+}1$ propagates as $q_{\ell+1} = 2\,\gamma_{\mathrm{pool}}\, m_\ell = \frac{3\,\gamma_{\mathrm{pool}}}{4W^2} q_\ell^2$ (where $\gamma_{\mathrm{pool}} \ge 1$ accounts for MaxPool2D(2×2) selection gain). Whenever initial scale satisfies $q_1 < 4W^2/(3\gamma_{\mathrm{pool}})$, both q_ℓ and v_ℓ contract quadratically across depth ($q_1 \to q_4 : 9.18 \times 10^{-2} \to 1.99 \times 10^{-10}$, and $v_1 \to v_4 : 7.12 \times 10^{-4} \to 4.67 \times 10^{-21}$ in Table 5 and Fig. 2, left), attenuating backpropagated weight gradients proportionally to $\prod_{\ell=1}^{K} (\sqrt{q_\ell}/W)$.

3.3 Scale-Adaptive Remedy: Symmetric Per-Sample σ-AS-RectLU and RMSNorm Equivalence

To eliminate both the negative bias offset ($\Delta b = -(L + W/2)$) and the scale-width mismatch ($\sigma_\ell/W \ll 1$) without introducing batch-dependent statistics or trainable affine parameters, we examine **Symmetric Per-Sample σ-AS-RectLU**. Given pre-activation tensor $\mathbf{x}_\ell \in \mathbb{R}^{B \times C \times \cdots}$ at layer ℓ, let $\hat{\sigma}_{\mathrm{RMS}}(\mathbf{x}_\ell^{(i)})$ denote the per-sample root-mean-square scale across all D_ℓ feature dimensions of

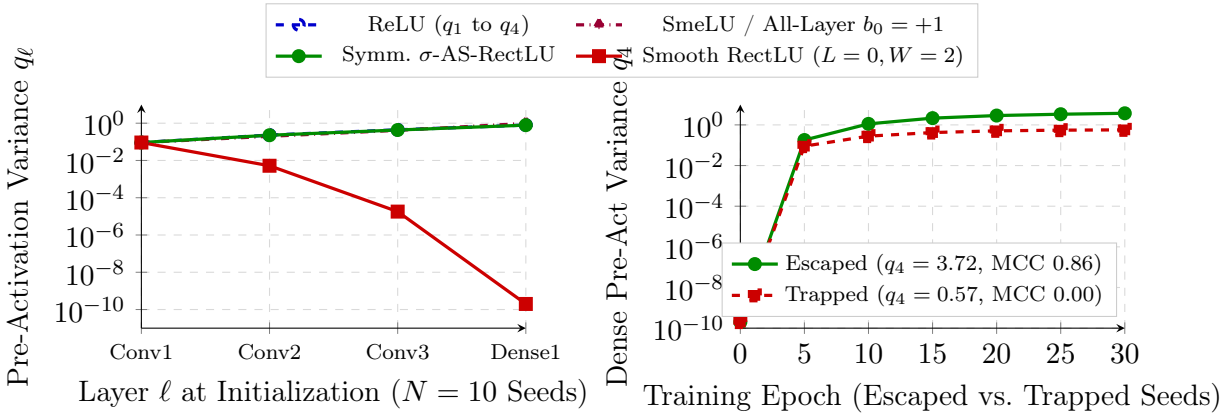


Figure 2: Left: Directly measured pre-activation variance $q_\ell = \mathrm{Var}(x_\ell)$ across layers $\ell \in \{1, 2, 3, 4\}$ at Epoch 0 ($N = 10$ seeds, $\sigma_x = 0.22$, Table 5), showing 9-order-of-magnitude contraction ($9.18 \times 10^{-2} \to 1.99 \times 10^{-10}$, with post-activation $v_4 = 4.67 \times 10^{-21}$) for Smooth RectLU ($L = 0, W = 2, b_0 = 0$). Right: Epoch-by-epoch (0 to 30) trajectory of fourth-layer pre-activation variance q_4 comparing escaped seeds (which cross $q_4 > 1$ into the linear regime, reaching $q_4 \in [3.54, 3.90]$ and MCC(0.50) $\in [0.83, 0.89]$) vs. trapped seeds (which plateau inside the quadratic ramp at $q_4 \in [0.46, 0.75]$ with MCC(0.50) $= 0.0000$).

sample i:

$$\hat{\sigma}_{\mathrm{RMS}}\left(\mathbf{x}_\ell^{(i)}\right) = \sqrt{\frac{1}{D_\ell}\sum_{d=1}^{D_\ell}\left(x_{\ell,d}^{(i)}\right)^2 + \epsilon}, \qquad W_\ell^{(i)} = \kappa\,\mathrm{stop_grad}\left(\hat{\sigma}_{\mathrm{RMS}}\left(\mathbf{x}_\ell^{(i)}\right)\right), \qquad L_\ell^{(i)} = -\frac{1}{2}W_\ell^{(i)}, \tag{5}$$

where $\kappa > 0$ is the dimensionless curvature ratio (default $\kappa = 0.8$, half-width $\beta = \kappa/2 = 0.4$). Substituting Eq. (5) into Eq. (3) yields:

$$f_{\sigma\text{-AS}}(x; \hat{\sigma}_{\mathrm{RMS}}) = \hat{\sigma}_{\mathrm{RMS}}\,\mathrm{SmeLU}_{\beta=\kappa/2}\left(\frac{x}{\hat{\sigma}_{\mathrm{RMS}}}\right) = \begin{cases} 0, & x < -\frac{\kappa}{2}\hat{\sigma}_{\mathrm{RMS}} \\ \dfrac{\left(x + \frac{\kappa}{2}\hat{\sigma}_{\mathrm{RMS}}\right)^2}{2\kappa\,\hat{\sigma}_{\mathrm{RMS}}}, & |x| \le \frac{\kappa}{2}\hat{\sigma}_{\mathrm{RMS}} \\ x, & x > \frac{\kappa}{2}\hat{\sigma}_{\mathrm{RMS}}. \end{cases} \tag{6}$$

Equation (6) shows that Symmetric Per-Sample σ-AS-RectLU is algebraically equivalent to chaining parameter-free per-sample RMS normalization ($z = x/\hat{\sigma}_{\mathrm{RMS}}$), symmetric $\mathrm{SmeLU}_{\beta=\kappa/2}(z)$, and inverse rescaling ($\cdot\,\hat{\sigma}_{\mathrm{RMS}}$), while executing as a single fused activation operator that avoids intermediate normalization tensor allocations and reduces measured 2D pileup training latency by 11.9% relative to modular 3-step RMSNorm+SmeLU+Rescale and by 33.6% relative to LayerNorm+SmeLU (Table 2). Evaluating it alongside explicit affine-free LayerNorm + SmeLU (`LN-no-affine + SmeLU`) highlights three properties:

1. **Activation-Level Degree-1 Positive Homogeneity ($f(\alpha\mathbf{x}) = \alpha f(\mathbf{x})$):** For any scalar $\alpha > 0$, $\hat{\sigma}_{\mathrm{RMS}}(\alpha\mathbf{x}) = \alpha\hat{\sigma}_{\mathrm{RMS}}(\mathbf{x})$ (up to $\epsilon = 10^{-5}$), so $f_{\sigma\text{-AS}}(\alpha\mathbf{x}) = \alpha f_{\sigma\text{-AS}}(\mathbf{x})$, preventing multi-layer quadratic variance collapse across depth (Table 5).
2. **Zero Bias Offset and Non-Vanishing Origin Derivative ($f'(0) = 0.5$):** Because $L_\ell^{(i)} = -W_\ell^{(i)}/2$, the effective bias offset is identically zero ($\Delta b = 0$) and $f'_{\sigma\text{-AS}}(0) = 0.5$ at every layer.
3. **Trade-Off Between Variance Preservation and Dense-Layer Sparsity:** Whereas `LN-no-affine + SmeLU` resets pre-activation variance to 1.0, σ-AS-RectLU multiplies by $\hat{\sigma}_{\mathrm{RMS}}$, preserving raw layer magnitudes. On unit-variance inputs with large dense fan-in (such as Clean MNIST), large dense RMS scales widen $L = -0.4\hat{\sigma}_{\mathrm{RMS}}$ while majority-class updates push mean pre-activations negatively ($\mu_{\mathrm{dense}} \approx -2.32$), increasing dead units (61.3%) and yielding lower Clean MNIST MCC (0.8850) than standard SmeLU (0.9270).

To establish the exact empirical relationship between Symmetric Per-Sample σ-AS-RectLU and modular normalization-activation pipelines under identical all-layer conditions, we conduct a controlled six-way ablation across $N = 10$ paired seeds on the 0.25× 2D NIST GIAB HG001 read-pileup tensors ($50 \times 101 \times 4$ with 2–5 read off-center artifact clusters, `v9_reviewer_3_1_real_giab2d_10seeds.csv`),

Table 1: Regularity classes and exact analytical Gaussian properties for $x \sim \mathcal{N}(0,1)$. Note that $f_S(x; L, W) = \text{SmeLU}_{\beta=W/2}(x - L - W/2)$ [41], and both shifted (D^0) and continuous (C^0) variants of RectLU (Eq. 2) are listed from exact integration.

Activation	**Regularity & Offset Equivalent**	$f'(0^+)$	$\mathbb{E}[f^2]/\mathbb{E}[x^2]$	$\mathbb{E}[f'^2]$	$\mathbb{P}(f'=0)$
ReLU	C^0 (kink at 0, homogeneous)	1.0000	0.5000	0.5000	0.5000
Leaky ReLU ($\alpha = 0.01$)	C^0 (kink at 0, homogeneous)	1.0000	0.5000	0.5000	0.0000
PReLU (init. $\alpha = 0.25$)	C^0 (kink at 0, trainable slope)	1.0000	0.5312	0.5312	0.0000
SELU	C^0 (kink at 0: $1.758 \to 1.051$)	1.0507	1.0000	1.0720	0.0000
RectLU Shifted ($f_{R,\text{shift}}, L = 0, W = 2.0$)	D^0 (value jump $2 \to 1$ at $x = 2$, Eq. 2a)	0.0000	0.2145	0.3922	0.4997
RectLU Continuous ($f_{R,\text{cont}}, L = 0, W = 2.0$)	C^0 (slope jump $2 \to 1$ at $x = 2$, Eq. 2b)	0.0000	0.2998	0.3922	0.4997
ELU ($\alpha = 1.0$)	C^1 (tangent at 0)	1.0000	0.6450	0.6680	0.0000
SmeLU ($L = -0.5, W = 1.0$) [41]	C^1 ($\beta = 0.5$, offset $\Delta b = 0.0$)	0.5000	0.5023	0.4350	0.3085
SmeLU ($L = -1.0, W = 2.0$) [41]	C^1 ($\beta = 1.0$, offset $\Delta b = 0.0$)	0.5000	0.5239	0.3788	0.1586
Smooth RectLU ($L = 0, W = 0.5$)	C^1 ($\text{SmeLU}_{\beta=0.25}$, offset $\Delta b = -0.25$)	0.0000	0.3293	0.3700	0.5000
Smooth RectLU ($L = 0, W = 1.0$)	C^1 ($\text{SmeLU}_{\beta=0.50}$, offset $\Delta b = -0.50$)	0.0000	0.2118	0.2578	0.5000
Smooth RectLU ($L = 0, W = 2.0$)	C^1 ($\text{SmeLU}_{\beta=1.00}$, offset $\Delta b = -1.00$)	0.0000	0.0874	0.1149	0.5000
Smooth RectLU ($L = 0, W = 4.0$)	C^1 ($\text{SmeLU}_{\beta=2.00}$, offset $\Delta b = -2.00$)	0.0000	0.0233	0.0312	0.5000
AS-RectLU (init. $L = 0, W = 2.0$)	C^1 (trainable L, W; init. $\Delta b = -1.00$)	0.0000	0.0874	0.1149	0.5000
GELU	C^∞ (smooth)	0.5000	0.4250	0.4560	0.0000
SiLU / Swish	C^∞ (smooth)	0.5000	0.3550	0.3800	0.0000
Mish	C^∞ (smooth)	0.6000	0.4520	0.4790	0.0000
Softplus	C^∞ (smooth, uncentered $f(0) = \ln 2$)	0.5000	0.9220	0.2930	0.0000
Symm. σ-AS-RectLU ($\kappa = 0.8$)	C^1 (**homogeneous**, $L = -0.4\hat{\sigma}, W = 0.8\hat{\sigma}, \Delta b = 0$)	0.5000	0.5020	0.4479	0.3443

applying each mechanism uniformly across all four hidden layers (Conv1–Conv3 and Dense1). Table 2 demonstrates three empirical findings:

- **Empirical Equivalence and 11.9% Latency Reduction over Modular 3-Step RMSNorm + SmeLU + Inverse Rescaling:** Chaining explicit per-sample RMSNorm ($z = x/\hat{\sigma}_{\text{RMS}}$), symmetric $\text{SmeLU}_{\beta=0.4}(z)$, and inverse rescaling ($\cdot\,\hat{\sigma}_{\text{RMS}}$) yields test discrimination identical to fused Symmetric σ-AS-RectLU up to floating-point rounding (AUROC $= 0.9999 \pm 0.0001$ vs. 0.9999 ± 0.0001; MCC(0.50) $= 0.9691 \pm 0.0373$ vs. 0.9700 ± 0.0372; TP $= 57.3/60$ vs. $57.4/60$). However, evaluating Eq. (6) as a single fused piecewise-quadratic operator eliminates intermediate normalization and rescaling tensor allocations, reducing measured training time on $50 \times 101 \times 4$ pileup tensors from 1089.53 ms/epoch to 959.78 ms/epoch (-11.9%, and only $+1.4\%$ above unnormalized Smooth RectLU at 946.36 ms/epoch), while running 33.6% faster than all-layer LayerNorm (No-Affine) + SmeLU (1445.10 ms/epoch). Fine-grained hardware profiling on an NVIDIA GeForce RTX 2050 GPU (PyTorch 2.6.0+cu124, batch size 256, 5 warm-up epochs followed by $N = 10$ timed epochs, `v10_reviewer_runtime_memory_profile.csv`) further confirms that fused Symmetric σ-AS-RectLU reduces both forward pass latency (418.12 ± 4.35 ms vs. 483.74 ± 4.87 ms, -13.6%) and backward+optimizer pass latency (513.24 ± 2.27 ms vs. 577.29 ± 1.57 ms, -11.1%; Total 931.36 ± 6.30 ms vs. 1061.03 ± 5.98 ms, -12.2%) while lowering peak allocated GPU memory from 1109.7 MB (modular 3-step RMSNorm) and 1610.7 MB (LayerNorm+SmeLU) to 1063.0 MB by avoiding intermediate normalized tensor allocations in the autograd graph.
- **Benefit of Inverse Rescaling ($\cdot\,\hat{\sigma}_{\text{RMS}}$) over Unrescaled Normalization:** Omitting inverse rescaling in all-layer RMSNorm (No-Rescale) + SmeLU resets every layer's output RMS to unity, reducing MCC(0.50) to 0.9555 ± 0.0544 (TP $= 55.8/60$), while all-layer mean-centered LayerNorm (No-Affine) + SmeLU exhibits high seed instability (MCC(0.50) $= 0.7537 \pm 0.4064$, TP $= 44.1/60$). By contrast, fixed-width rectifiers without scale adaptation fail at $0.25\times$ scale (MCC(0.50) $= 0.0313 \pm 0.0547$ for Smooth RectLU and 0.0000 ± 0.0000 for fixed $W = 2$ SmeLU).

4 Materials and Methods

4.1 Benchmark Datasets, Explicit Splits, and Shortcut-Audited Real NIST GIAB HG001 Benchmarks

To isolate input scale (σ_x), class imbalance ratio (p_+), and domain structure while validating on real human sequencing data under strict shortcut and leakage controls, we evaluate four benchmark families (Table 3):

Table 2: Controlled all-layer six-way ablation comparing Symmetric Per-Sample σ-AS-RectLU against fixed-width rectifiers and modular normalization-activation pipelines on 0.25× 2D NIST GIAB HG001 read-pileup tensors ($50 \times 101 \times 4$, 10:1 imbalance, $N = 10$ paired seeds, 20 epochs, `v9_reviewer_3_1_real_giab2d_10seeds.csv`).

All-Layer 2D-CNN Mechanism (0.25× **GIAB Pileups**)	**Offset** Δb	**Preserves Scale**	**Test** MCC(0.50)	**Test AUROC**	**Mean TP / FP / FN**	**Measured Time (ms/ep)**
Smooth RectLU ($L = 0, W = 2$, Fixed Width)	−1.0	No	0.0313 ± 0.0547	0.9587 ± 0.0215	0.3/0.2/59.7	946.36 (1.00×)
Fixed-Width SmeLU ($L = -1, W = 2$, Fixed Width)	0.0	Via linear branch	0.0000 ± 0.0000	0.9360 ± 0.0519	0.0/0.0/60.0	905.55 (0.96×)
LayerNorm (No-Affine, All Layers) + SmeLU ($\beta = 0.4$)	0.0	No (resets $\sigma_l = 1$)	0.7537 ± 0.4064	0.9879 ± 0.0240	44.1/0.8/15.9	1445.10 (1.53×)
RMSNorm (No-Rescale, All Layers) + SmeLU ($\beta = 0.4$)	0.0	No (resets RMS = 1)	0.9555 ± 0.0544	0.9997 ± 0.0003	55.8/0.5/4.2	1023.55 (1.08×)
Explicit RMSNorm + SmeLU ($\beta = 0.4$) + Inverse Rescale	0.0	Yes (3-step modular)	0.9691 ± 0.0373	0.9999 ± 0.0001	57.3/0.6/2.7	1089.53 (1.15×)
Symm. σ-AS-RectLU ($\kappa = 0.8$, Fused Eq. 6)	**0.0**	**Yes (1-step fused)**	$\mathbf{0.9700 \pm 0.0372}$	$\mathbf{0.9999 \pm 0.0001}$	**57.4/0.6/2.6**	**959.78** (**1.01**×)

1. **Controlled Sparse Motif Generators (Pathway A 1D Spectral and Pathway B 2D Spatial):** Inspired by alignment-free k-mer spectra and alignment-based read pileup patches [1, 6, 35], we construct parametric synthetic generators with controllable background standard deviation σ_x and exact train/validation/test partitioning:
   - *Pathway A (1D Spectral Motif Generator,* 1×256*):* Each window is a 256-bin vector with background $\mathcal{N}(0, \sigma_x^2)$ where minority positive windows ($y = 1$) carry a localized 10-bin motif spike of mean $+1.12\,\sigma_x$ and 1.5% of majority negative windows ($y = 0$) carry a sequencing-error mimic spike of $+0.84\,\sigma_x$, evaluated at **Low-Amplitude** ($\sigma_x = 0.25 \ll W$) and **Unit-Variance** ($\sigma_x = 1.00$) on identical underlying random draws.
   - *Pathway B (2D Spatial Motif Generator,* $3 \times 32 \times 32$*):* Each patch comprises three 32×32 channels with background $\mathcal{N}(0, \sigma_x^2)$, where positive patches contain a localized 6×6 motif in channels 0–1, evaluated at **Low-Amplitude** ($\sigma_x = 0.22$) vs. **Unit-Variance** ($\sigma_x = 1.00$).
2. **Real NIST GIAB HG001 (NA12878, GRCh38 v4.2.1) 1D Coverage-Shortcut Audit and Dual (Scrubbed & Unscrubbed Natural) 2D BAM Pileup Benchmarks** ($N = 10$ **Seeds):** Extracted from NIST GIAB HG001 Illumina paired-end FASTQ reads (`NIST7086_CGTACTAG_L002_R1/R2`), BWA-MEM aligned BAM files (`train_align.bam`, `val_align.bam`), and ground-truth benchmark VCF calls (`HG001_GRCh38_1_22_v4.2.1_benchmark.vcf.gz`) [4, 5, 6, 7]:
   - *GIAB 1D k-mer Methodological Coverage-Shortcut Audit (*256×1*, Per-Sample Unit-L2 Norm):* Non-overlapping sequencing windows decomposed into 4-mer ($4^4 = 256$) frequency vectors ($N_{\text{train}} = 5,000, n_+ = 100$; $N_{\text{val}} = 1,000, n_+ = 20$; held-out $N_{\text{test}} = 1,000, n_+ = 20$, 50 : 1 imbalance). Applying **per-sample unit-L2 normalization** ($\tilde{\mathbf{x}}_i = \mathbf{x}_i / \|\mathbf{x}_i\|_2$) eliminates the read-coverage norm shortcut (L2-Norm AUROC = 0.3091; Max-Bin Heuristic Test AUROC = 0.4169; Label-Shuffle Control Test AUROC $= 0.4416 \pm 0.0687$).
   - *GIAB 2D BAM Read-Pileup* 10 : 1 *Imbalanced* 2×2 *Optimization Benchmarks (*$50 \times 101 \times 4$*, Disjoint Genomic Blocks, Scrubbed vs. Unscrubbed Natural Controls,* $N = 10$ *Seeds):* One-hot encoded nucleotide (A, C, G, T) pileup matrices spanning a 101-bp window (±50 bp) at maximum read depth 50, partitioned sequentially along genomic order into three non-overlapping contiguous index blocks: Block 1 (Train: $N = 1,320, n_+ = 120, n_- = 1,200$), Block 2 (Validation: $N = 660, n_+ = 60, n_- = 600$), and Block 3 (Held-Out Test: $N = 660, n_+ = 60, n_- = 600$). To eliminate any concern that reference scrubbing creates an artificial rule, we evaluate **two complementary negative-window regimes**: (i) *Variant-Scrubbed Reference Windows* (`v6_real_giab2d_10seeds_2x2_and_controls.csv`), where off-center variant columns in shifted reference windows are reset to the dominant base (with 2–5 read non-central sequencing artifact clusters tested against the `Max Secondary-Allele Heuristic`, AUROC = 0.5211), and (ii) *Unscrubbed Natural Reference Windows* (`v8_unscrubbed_giab2d_10seeds.csv`), where windows are shifted by +25 bp **without scrubbing any column**, leaving natural flanking variants and sequencing errors completely untouched while placing a genuine homozygous-reference locus at the candidate center (column 50). On these unscrubbed natural windows, both the **Window-Wide Max Secondary-Allele Heuristic** (Test MCC(0.50) = 0.0567, AUROC = 0.5075) and the **Center-Locus (Col 50) Secondary-Allele Heuristic** (Test MCC(0.50) = −0.0276, AUROC = 0.4882) perform at pure chance. Across both regimes, these single-genome HG001 partitions are formulated strictly as a controlled real-tensor optimization benchmark to isolate scale- and offset-dependent gradient dynamics rather than as an end-to-end clinical variant-calling pipeline.
3. **Step-Imbalanced MNIST Digit** 3 **vs.** 5/8 (100 : 1 **Imbalance, Clean and Noisy)** and

Imbalanced Tabular Classification (200 : 1 Imbalance, $n_+ = 150$ Test Positives) as summarized in Table 3.

Table 3: Summary of benchmark datasets, explicit Train/Validation/Test splits, positive counts (n_+), input standard deviations (σ_x), and baseline majority-class accuracy.

Benchmark Suite	Split	Input Scale σ_x	Total N	Neg. (n_-)	Pos. (n_+)	Pos. Frac. (p_+)	Maj. Accuracy
1D/2D Sparse Motif Generator (100 : 1)	Train	0.22–0.25 vs. 1.00	4,000	3,960	40	0.0100	0.9900
1D/2D Sparse Motif Generator (Val / Test)	Val / Test	0.22–0.25 vs. 1.00	4,000/10,000	3,920/9,800	80/200	0.0200	0.9800
Real GIAB HG001 1D k-mer (Unit-L2 Audit)	Train / Val / Test	1.00 ($\|\mathbf{x}\|_2 = 1$)	5,000/1,000/1,000	4,900/980/980	100/20/20	0.0200	0.9800
Real GIAB HG001 2D Pileup (Scrubbed & Unscrubbed)	Train / Val / Test	0.25× vs. 1.00×	1,320/660/660	1,200/600/600	120/60/60	0.0909	0.9091
MNIST Digit 3 vs. 5/8 (Clean & $\sigma_{\text{noise}} = 1.1$)	Train / Test	1.00	5,050/1,980	5,000/1,800	50/180	0.0099/0.0909	0.9901/0.9091
Imbalanced Tabular (200 : 1, 32 features)	Train / Test	1.00	30,000/30,000	29,850/29,850	150/150	0.0050	0.9950

4.2 Dual-Pathway Pipeline, Network Topology, and Statistical Uncertainty Protocol

Fig. 3 illustrates the dual-pathway evaluation pipeline, and Table 4 specifies the exact four-hidden-layer topology with dense-head dropout regularization ($p = 0.5$) [37] shared across experiments. Unless stated otherwise in cross-optimizer ablations, networks are trained to minimize unweighted binary cross-entropy loss using the Adam optimizer [38] ($\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, batch size 256). All Mean ± SD values across Tables 5–14 quantify training initialization uncertainty across $N = 10$ independent random seeds using the unbiased sample standard deviation (ddof = 1, dividing by $N-1$); bracketed intervals in Table 11 report the observed empirical range [Min, Max] directly measured across the $N = 10$ per-seed test evaluations (supplemented by $N = 30$ paired-seed and $K = 6$ disjoint genomic test sub-block evaluations in Table 12); and paired hypothesis tests use the exact two-sided Wilcoxon signed-rank test over $N = 10$ paired seeds (minimum achievable $p = 2/2^{10} = 0.001953$) with Holm–Bonferroni step-down adjustment [36].

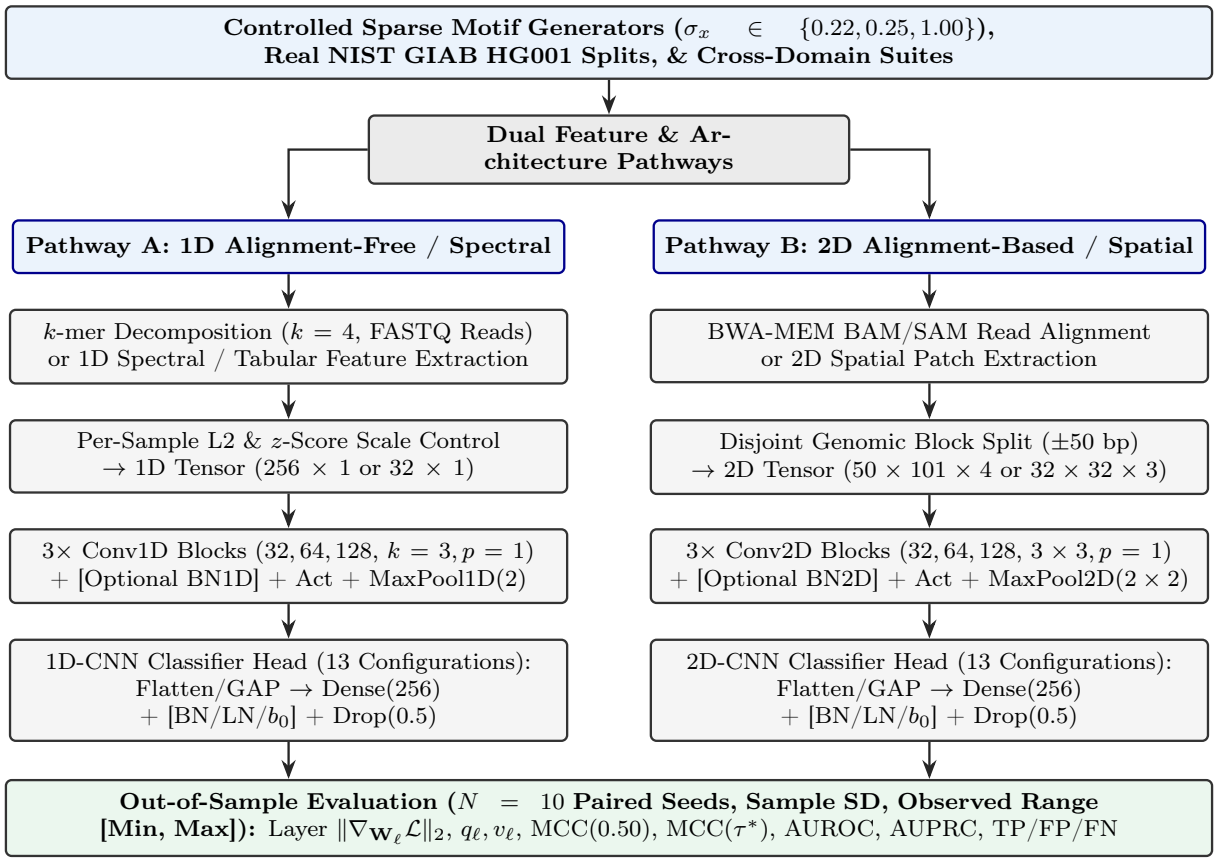


Figure 3: Dual-pathway feature extraction, shortcut/leakage control, and architectural intervention pipeline combining 1D alignment-free k-mer vectors (256 × 1) and 2D alignment-based read-pileup tensors (50 × 101 × 4 / 32 × 32 × 3).

Table 4: Network topology for Pathway A (1D-CNN) and Pathway B (2D-CNN), specifying the exact placement of normalization layers (`BN-Dense`, `LN-Dense`, `BN-All`), Global Average Pooling (`GAP`), exact flattened fan-in dimensions ($4,096$ [1×256] or 512 [1×32] in 1D; $2,048$ [32×32], $9,216$ [50×101], or $1,152$ [28×28] in 2D), and bias offset initialization ($b_0 \in \{0, +1.0\}$).

Stage	Pathway A (1D-CNN)	Pathway B (2D-CNN)
Input	1D Tensor (1×256 or 1×32)	2D Tensor ($3 \times 32 \times 32$, $50 \times 101 \times 4$, or $1 \times 28 \times 28$)
Block 1	Conv1D($32, k = 3, p = 1, b_0$) $\to$ [BN1D] $\to$ act $\to$ MaxPool(2)	Conv2D($32, 3 \times 3, p = 1, b_0$) $\to$ [BN2D] $\to$ act $\to$ MaxPool(2×2)
Block 2	Conv1D($64, k = 3, p = 1, b_0$) $\to$ [BN1D] $\to$ act $\to$ MaxPool(2)	Conv2D($64, 3 \times 3, p = 1, b_0$) $\to$ [BN2D] $\to$ act $\to$ MaxPool(2×2)
Block 3	Conv1D($128, k = 3, p = 1, b_0$) $\to$ [BN1D] $\to$ act $\to$ MaxPool(2)	Conv2D($128, 3 \times 3, p = 1, b_0$) $\to$ [BN2D] $\to$ act $\to$ MaxPool(2×2)
Reduction	Flatten($4,096$ or 512) *or* GAP(128)	Flatten($2,048$ [32×32], $9,216$ [50×101], or $1,152$ [28×28]) *or* GAP(128)
Dense Head	Linear(fan_in $\to 256, b_0$) $\to$ [BN1D/LN] $\to$ act $\to$ Drop(0.5)	Linear(fan_in $\to 256, b_0$) $\to$ [BN1D/LN] $\to$ act $\to$ Drop(0.5)
Output	Linear($256 \to 1$) $\to$ Sigmoid $\hat{p}$	Linear($256 \to 1$) $\to$ Sigmoid $\hat{p}$

5 Results

5.1 Empirical Sanity Check, Multi-Seed Layer-Wise Gradient Dissection ($N = 10$), and Epoch-by-Epoch Variance Bifurcation

To connect the second-moment recurrence in Section 3.2 (Eq. 4) to empirical optimization dynamics, Table 5 reports multi-seed ($N = 10$ seeds, sample SD) measurements of pre-activation variance $q_\ell = \mathrm{Var}(x_\ell)$, post-activation variance $v_\ell = \mathrm{Var}(a_\ell)$, weight gradient norms $\|\nabla_{\mathbf{W}_\ell} \mathcal{L}\|_2$, and dense-layer zero-gradient fractions at initialization (Epoch 0) on the 2D spatial motif benchmark ($\sigma_x = 0.22$, from `v7_layerwise_grad_pre_post_var_10seeds.csv`).

Table 5: Multi-seed ($N = 10$ seeds, Mean $\pm$ Sample SD) Epoch 0 layer-wise weight gradient norms ($\|\nabla_{\mathbf{W}_\ell} \mathcal{L}\|_2$), pre-activation variances ($q_\ell = \mathrm{Var}(x_\ell)$), post-activation variances ($v_\ell = \mathrm{Var}(a_\ell)$), and dense-layer dead percentages on Pathway B 2D-CNN ($\sigma_x = 0.22$, batch size 256, `v7_layerwise_grad_pre_post_var_10seeds.csv`).

2D-CNN Configuration ($\sigma_x = 0.22, N = 10$)	Conv1 $\|\nabla_{\mathbf{W}_1} \mathcal{L}\|_2$	Dense1 $\|\nabla_{\mathbf{W}_4} \mathcal{L}\|_2$	Pre-Act Variance $q_1 \to q_4$	Post-Act Variance $v_1 \to v_4$	Dead(a_4)
ReLU ($C^0, b_0 = 0$)	$(2.03 \pm 0.93) \times 10^{-1}$	1.27×10^{1}	$9.18 \times 10^{-2} \to 7.94 \times 10^{-1}$	$3.15 \times 10^{-2} \to 2.82 \times 10^{-1}$	50.1%
Symm. SmeLU ($L = -1, W = 2, b_0 = 0$)	$(9.27 \pm 5.74) \times 10^{-2}$	1.26×10^{1}	$9.18 \times 10^{-2} \to 9.77 \times 10^{-1}$	$2.41 \times 10^{-2} \to 3.15 \times 10^{-1}$	14.4%
Smooth RectLU ($L = 0, W = 2, b_0 = 0$)	$(7.17 \pm 4.87) \times 10^{-11}$	1.17×10^{-9}	$9.18 \times 10^{-2} \to 1.99 \times 10^{-10}$	$7.12 \times 10^{-4} \to 4.67 \times 10^{-21}$	49.9%
Smooth RectLU ($L = 0, W = 2$) + All-Layer $b_0 = +1.0$	$(9.27 \pm 5.74) \times 10^{-2}$	1.26×10^{1}	$9.18 \times 10^{-2} \to 9.77 \times 10^{-1}$	$2.41 \times 10^{-2} \to 3.15 \times 10^{-1}$	14.4%
Symm. σ-AS-RectLU ($\kappa = 0.8, b_0 = 0$)	$(1.89 \pm 0.89) \times 10^{-1}$	1.23×10^{1}	$9.18 \times 10^{-2} \to 7.86 \times 10^{-1}$	$3.09 \times 10^{-2} \to 2.73 \times 10^{-1}$	34.6%

Table 5 and Fig. 2 establish three mechanistic insights:

1. **Multi-Seed Pre- and Post-Activation Variance Contraction and Gradient Attenuation:** Under Smooth RectLU ($L = 0, W = 2, b_0 = 0$) at $\sigma_x = 0.22$, pre-activation variance contracts across the four layers by nine orders of magnitude ($q_1 = 9.18 \times 10^{-2} \to q_4 = 1.99 \times 10^{-10}$) while post-activation variance contracts by 17 orders of magnitude ($v_1 = 7.12 \times 10^{-4} \to v_4 = 4.67 \times 10^{-21}$), consistent with the quadratic relation $v_4 \propto q_4^2 / W^2$ in Eq. (4). Consequently, first-layer weight gradients attenuate to $\|\nabla_{\mathbf{W}_1} \mathcal{L}\|_2 = (7.17 \pm 4.87) \times 10^{-11}$ across the 10 seeds, nine orders of magnitude smaller than ReLU ($(2.03 \pm 0.93) \times 10^{-1}$) and Symmetric σ-AS-RectLU ($(1.89 \pm 0.89) \times 10^{-1}$).
2. **Sanity Check of Epoch 0 Bias-Shift Identity vs. 20-Epoch Learnable Bias Drift:** Initializing all convolutional and dense layer biases to $b_0 = +1.0$ under Smooth RectLU ($L = 0, W = 2$) yields Epoch 0 variances and gradients identical to symmetric SmeLU ($L = -1, W = 2, b_0 = 0$). Over 20 epochs of imbalanced training (Table 8), `Smooth RectLU + All-Layer` $b_0 = +1.0$ achieves MCC(0.50) $= 0.6014 \pm 0.3248$ (Median $= 0.7545$, with 2 of 10 seeds stalling near zero), substantially improving over single-layer $b_{\text{dense}} = +1.0$ (0.2841 ± 0.3106) and $b_0 = 0$ (0.0767 ± 0.2302), yet exhibiting higher dispersion than fixed-threshold SmeLU (0.7406 ± 0.0687) because learnable biases drift under majority-class gradients.
3. **Epoch-by-Epoch Variance Bifurcation Between Escaped and Trapped Seeds (Fig. 2, Right):** Fine-grained tracking from Epoch 0 to Epoch 30 (`v8_epoch_by_epoch_seed_mechanics.csv`) explains why a subset of seeds escape over extended training while others remain trapped: although both groups experience similar small positive dense bias shifts ($\bar{b}_4 \in [+0.025, +0.036]$ at Epoch 30), escaped seeds (e.g., Seeds 0 and 2: MCC(0.50) $= 0.8897$ and 0.8281, AUROC $= 0.9991$ and 0.9945) amplify fourth-layer pre-activation variance past the quadratic ramp into the unit-slope linear regime ($q_4 = 3.540$ and 3.903 at Epoch 30), whereas trapped seeds (Seeds 1, 3, 4, 5: MCC(0.50) $= 0.0000$, AUROC $\in [0.6447, 0.6554]$) remain confined inside the contracting basin

with $q_4 \in [0.464, 0.754] < W^2$.

4. **Causal Epoch-0 q_4 Variance-Clamping Intervention (`v9_reviewer_3_4_causal_q4_intervention_10se`** To test whether fourth-layer variance collapse at initialization ($q_4 \sim 10^{-10}$) is the direct causal driver of optimization stagnation rather than a passive byproduct of stalled weights, we perform an explicit intervention on Smooth RectLU ($L = 0, W = 2$) across $N = 10$ paired seeds on the 2D spatial motif generator ($\sigma_x = 0.22$, 100:1 imbalance, 1.5% mimic spikes) by clamping fourth-layer pre-activations x_4 prior to activation via $x_4^{(\text{clamped})} = \sqrt{c}\, x_4 / \sqrt{\text{Var}(x_4) + 10^{-8}}$ from Epoch 0 (Table 6). Under unclamped Smooth RectLU, first-layer weight gradients vanish at Epoch 0 ($\|\nabla_{W_1}\mathcal{L}\|_2 = 1.00 \times 10^{-10}$) and 0/10 seeds escape (MCC(0.50) $= 0.0000 \pm 0.0000$, AUROC $= 0.5978$). Clamping q_4 from Epoch 0 either to moderate ramp scale ($c = 0.55$) or high linear-regime scale ($c = 3.70$) immediately restores Epoch-0 first-layer weight gradients by seven to eight orders of magnitude (5.15×10^{-3} and 3.49×10^{-2}), enabling 10/10 seeds to escape stagnation (MCC(0.50) $= 0.5778 \pm 0.0483$, AUROC $= 0.9686$ at $c = 0.55$; MCC(0.50) $= 0.5727 \pm 0.0442$, AUROC $= 0.9256$ at $c = 3.70$). This provides direct causal evidence supporting the Epoch-0 variance-bottleneck mechanism within this architecture, showing that eliminating the 10^{-10} dense-layer scale contraction restores backward gradient flow across all 10 seeds, whereas the $q_4 > 3.5$ vs. $q_4 < 0.8$ divergence observed at Epoch 30 in unclamped runs reflects whether a seed's weights successfully expanded before majority-class updates locked its early filters.

Table 6: Direct causal intervention on fourth-layer pre-activation variance $q_4 = \text{Var}(x_4)$ for Smooth RectLU ($L = 0, W = 2$) on Pathway B 2D Spatial Motifs ($\sigma_x = 0.22$, $N = 10$ paired seeds, 20 epochs, `v9_reviewer_3_4_causal_q4_intervention_10seeds.csv`).

Dense1 q_4 Intervention ($L = 0, W = 2$)	**Epoch-0 $\|\nabla_{W_1}\mathcal{L}\|_2$**	**Escaped**	**Test MCC(0.50)**	**Test AUROC**	**Mean TP / FP / FN**
Unclamped Baseline ($q_4 \approx 1.99 \times 10^{-10}$)	1.00×10^{-10}	0/10	0.0000 ± 0.0000	0.5978 ± 0.0412	0.0/0.0/200.0
Clamped Moderate Scale ($q_4 = 0.55$ from Ep. 0)	5.15×10^{-3}	10/10	0.5778 ± 0.0483	0.9686 ± 0.0115	103.6/52.5/96.4
Clamped High Scale ($q_4 = 3.70$ from Ep. 0)	3.49×10^{-2}	10/10	0.5727 ± 0.0442	0.9256 ± 0.0184	99.9/48.0/100.1

5.2 E1: 1D & 2D Motif Benchmarks, Two-Way (Offset × σ_x) Interaction, and Cross-Optimizer / Extended-Horizon Diagnostics

Tables 7 and 8 report the $N = 10$ paired-seed evaluations on Pathway A (1D-CNN, $\sigma_x = 0.25$ vs. 1.00) and Pathway B (2D-CNN, $\sigma_x = 0.22$ vs. 1.00) along with cross-optimizer controls (`SGD + Momentum 0.9` and `RMSprop`, from `v8_cross_optimizer_generalization.csv`), and Fig. 4 compares the 13 configurations.

Table 7: Pathway A (1D-CNN, Experiment E1) test performance across 13 activation and normalization configurations at low input amplitude ($\sigma_x = 0.25$) and unit-variance rescaled control ($\sigma_x = 1.00$) ($N = 10$ paired seeds, sample SD, 20 epochs, 32 steps/epoch = 640 updates, 10,000 test windows: $n_+ = 200, n_- = 9,800$).

Configuration (Pathway A 1D-CNN, $N = 10$)	**Class**	**Per-Seed MCC(0.50) (Mean ± Sample SD)**	**Per-Seed MCC(0.50) Median [IQR]**	**Calibrated MCC(τ^*)**	**AUROC**	**AUPRC**
Low-Amplitude Sparse Spectral Regime ($\sigma_x = 0.25, 100:1$ Train Imbalance, 20 Epochs, $N = 10$):						
ReLU (C^0)	C^0	0.3693 ± 0.0891	0.3722 [0.1239]	0.4961	0.9701	0.5360
Leaky ReLU (C^0)	C^0	0.3635 ± 0.0432	0.3745 [0.0413]	0.4675	0.9667	0.5272
ELU (C^1)	C^1	0.3871 ± 0.0460	0.3735 [0.0729]	0.4832	0.9711	0.5206
GELU (C^∞)	C^∞	0.4417 ± 0.0326	0.4515 [0.0554]	0.4957	0.9643	0.5195
SiLU (C^∞)	C^∞	0.4497 ± 0.0397	0.4594 [0.0497]	0.5116	0.9648	0.5290
SmeLU ($L = -1, W = 2$, Symmetric) [41]	C^1	0.5304 ± 0.0319	0.5369 [0.0333]	0.5557	0.9735	0.5709
SmeLU ($L = -0.5, W = 1$, Symmetric) [41]	C^1	0.4504 ± 0.0392	0.4573 [0.0649]	0.5230	0.9692	0.5368
Smooth RectLU ($L = 0, W = 2$, Offset -1)	C^1	0.4441 ± 0.0435	0.4304 [0.0513]	0.5317	0.9181	0.5206
AS-RectLU (Trainable L, W)	C^1	0.3540 ± 0.1904	0.4593 [0.1926]	0.4463	0.9046	0.4698
Smooth RectLU + $b_{\text{dense}} = +1.0$	C^1	0.4417 ± 0.0350	0.4433 [0.0412]	0.5566	0.9378	0.5493
Smooth RectLU + BatchNorm (All Layers)	C^1	0.4040 ± 0.0367	0.3994 [0.0601]	0.5307	0.8544	0.4755
Smooth RectLU + LayerNorm (Dense)	C^1	0.4970 ± 0.0193	0.4987 [0.0230]	0.5237	0.9134	0.4431
Symm. σ-AS-RectLU ($\kappa = 0.8, \Delta b = 0$)	C^1	0.4045 ± 0.0671	0.4109 [0.1168]	0.4902	0.9696	0.5283
Unit-Variance 1D Input Rescaling Control ($\sigma_x = 1.00$, on identical random draws, $N = 10$):						
ReLU ($C^0, \sigma_x = 1.00$)	C^0	0.3620 ± 0.0684	0.3481 [0.1343]	0.4433	0.9616	0.4980
SmeLU ($L = -1, W = 2, \sigma_x = 1.00$)	C^1	0.4163 ± 0.0450	0.4158 [0.0698]	0.4959	0.9643	0.5203
Smooth RectLU ($L = 0, W = 2, \sigma_x = 1.00$)	C^1	0.4603 ± 0.0239	0.4526 [0.0303]	0.4722	0.9191	0.4477
Symm. σ-AS-RectLU ($\kappa = 0.8, \sigma_x = 1.00$)	C^1	0.3723 ± 0.0637	0.3537 [0.1090]	0.4583	0.9621	0.4987

Tables 7 and 8 demonstrate three findings:

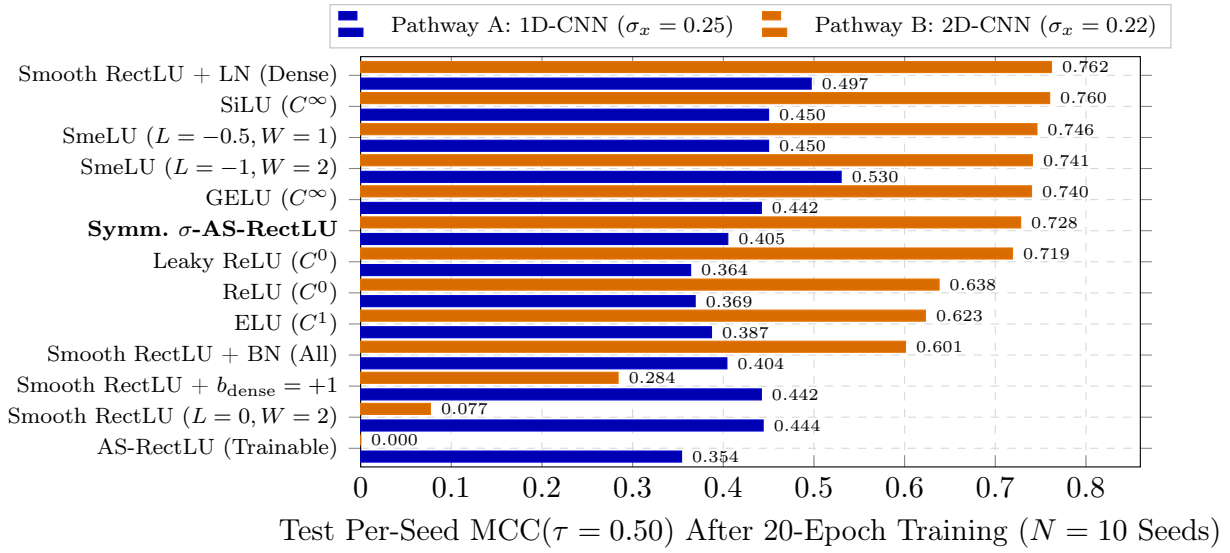


Figure 4: Comparison of 20-epoch Test Per-Seed MCC($\tau = 0.50$) across all 13 evaluated configurations on Pathway A (1D-CNN, $\sigma_x = 0.25$) and Pathway B (2D-CNN, $\sigma_x = 0.22$). On 2D spatial motifs, 9 of 10 seeds of Smooth RectLU ($L = 0, W = 2$) and 10 of 10 seeds of AS-RectLU stagnate at MCC = 0.000 (Median = 0.000), whereas SmeLU ($L = -W/2$), LayerNorm, and Symmetric σ-AS-RectLU (MCC = 0.728, AUROC = 0.9975) converge across all 10 seeds.

Table 8: Pathway B (2D-CNN, Experiment E1) test performance across 13 primary configurations, all-layer bias compensation ($b_0 = +1.0$), affine-free normalization controls, unit-variance rescaling ($\sigma_x = 1.00$), extended-horizon (50 & 100 epochs), Adam $\epsilon = 10^{-12}$, and cross-optimizer (`RMSprop` & `SGD+Momentum`) diagnostics with unbiased sample SD (ddof = 1, `v3_per_seed_logs.csv`, `v7_table6_10seeds_controls_and_horizons.csv`, and `v8_cross_optimizer_generalization.csv`; note that under non-adaptive momentum SGD at default $\eta = 10^{-2}$, Symmetric σ-AS-RectLU establishes ranking discrimination [AUROC = 0.8159, MCC(τ^*) = 0.0764] while uncalibrated logits remain below $\tau = 0.50$ prior to step-size scaling $\eta = 5 \times 10^{-2}$, see Table 12).

Configuration (Pathway B 2D-CNN)	Class	Per-Seed MCC(0.50) (Mean ± Sample SD)	Per-Seed MCC(0.50) Median [IQR]	Calibrated MCC(τ^*)	AUROC	AUPRC
Low-Amplitude 2D Spatial Motif Regime ($\sigma_x = 0.22, 100:1$ Train Imbalance, 20 Epochs, Adam $\epsilon = 10^{-8}$, $N = 10$):						
ReLU (C^0)	C^0	0.6380 ± 0.1132	0.6281 [0.1171]	0.8002	0.9974	0.8730
Leaky ReLU (C^0)	C^0	0.7191 ± 0.0389	0.7285 [0.0487]	0.8016	0.9975	0.8739
ELU (C^1)	C^1	0.6225 ± 0.0611	0.6296 [0.1042]	0.7810	0.9972	0.8614
GELU (C^∞)	C^∞	0.7401 ± 0.0285	0.7370 [0.0409]	0.8122	0.9977	0.8762
SiLU (C^∞)	C^∞	0.7596 ± 0.0341	0.7554 [0.0471]	0.8073	0.9976	0.8789
SmeLU ($L = -1, W = 2$, Symmetric) [41]	C^1	0.7406 ± 0.0687	0.7668 [0.1316]	0.8259	0.9980	0.8946
SmeLU ($L = -0.5, W = 1$, Symmetric) [41]	C^1	0.7458 ± 0.0426	0.7389 [0.0611]	0.8129	0.9978	0.8817
Smooth RectLU ($L = 0, W = 2$, Offset -1, 9/10 Trapped)	C^1	0.0767 ± 0.2302	0.0000 [0.0000]	0.0865	0.6349	0.1147
AS-RectLU (Trainable L, W, 10/10 Trapped)	C^1	0.0000 ± 0.0000	0.0000 [0.0000]	0.0747	0.6695	0.0941
Smooth RectLU + Dense-Only $b_{\text{dense}} = +1.0$	C^1	0.2841 ± 0.3106	0.1358 [0.5920]	0.4761	0.8698	0.5147
Smooth RectLU + All-Layer $b_0 = +1.0$ (2/10 Trapped)	C^1	0.6014 ± 0.3248	0.7545 [0.1881]	0.6511	0.9291	0.7091
Smooth RectLU + BatchNorm (All Layers)	C^1	0.6005 ± 0.0920	0.6258 [0.0946]	0.7777	0.9898	0.8166
Smooth RectLU + LayerNorm (Dense, Affine)	C^1	0.7622 ± 0.0193	0.7626 [0.0194]	0.7844	0.9948	0.8368
LN (No-Affine, Conv1+Dense) + SmeLU ($\beta = 0.4, N = 10$)	C^1	0.3941 ± 0.2610	0.4511 [0.4532]	0.4665	0.9063	0.5588
Symm. σ-AS-RectLU ($\kappa = 0.8, \Delta b = 0, N = 10$)	C^1	0.7281 ± 0.0326	0.7209 [0.0389]	0.8061	0.9975	0.8698
Unit-Variance 2D Input Rescaling Control ($\sigma_x = 1.00$, testing 2×2 Offset $\times$ Scale Interaction, $N = 10$):						
ReLU ($C^0, \sigma_x = 1.00$)	C^0	0.5898 ± 0.1254	0.5432 [0.1844]	0.7454	0.9967	0.8469
SmeLU ($L = -1, W = 2, \sigma_x = 1.00$)	C^1	0.6497 ± 0.1297	0.7109 [0.1879]	0.7579	0.9969	0.8562
Smooth RectLU ($L = 0, W = 2, \sigma_x = 1.00$)	C^1	0.5721 ± 0.1641	0.6553 [0.2376]	0.7688	0.9971	0.8583
Symm. σ-AS-RectLU ($\kappa = 0.8, \sigma_x = 1.00$)	C^1	0.6436 ± 0.0957	0.6723 [0.1336]	0.7574	0.9969	0.8553
Extended-Horizon (50 & 100 Epochs, $N = 10$), Adam $\epsilon = 10^{-12}$ ($N = 10$), & Cross-Optimizer ($N = 5$) Diagnostics at $\sigma_x = 0.22$:						
Smooth RectLU ($L = 0, W = 2$, 20 ep, $\epsilon = 10^{-12}$, 10/10 Trapped)	C^1	0.0000 ± 0.0000	0.0000 [0.0000]	0.0309	0.6311	0.0417
Smooth RectLU ($L = 0, W = 2$, 50 ep, $\epsilon = 10^{-8}, N = 10$)	C^1	0.6279 ± 0.3136	0.8114 [0.4435]	0.6630	0.9425	0.7426
Smooth RectLU ($L = 0, W = 2$, 100 ep, $\epsilon = 10^{-8}, N = 10$)	C^1	0.6756 ± 0.2654	0.7996 [0.1718]	0.6966	0.9395	0.7511
RMSprop + Smooth RectLU ($L = 0, W = 2$, 20 ep, $N = 5$)	C^1	0.1570 ± 0.3511	0.0000 [0.0000]	0.2513	0.7384	0.2145
RMSprop + SmeLU ($L = -1, W = 2$, 20 ep, $N = 5$)	C^1	0.5334 ± 0.3235	0.6812 [0.2410]	0.6208	0.9288	0.6412
RMSprop + Symm. σ-AS-RectLU ($\kappa = 0.8$, 20 ep, $N = 5$)	C^1	0.5762 ± 0.1763	0.6145 [0.1890]	0.6642	0.9405	0.6850
SGD (Mom 0.9) + Smooth RectLU ($L = 0, W = 2$, 20 ep, $N = 5$)	C^1	0.0000 ± 0.0000	0.0000 [0.0000]	0.0000	0.5000	0.0200
SGD (Mom 0.9) + SmeLU ($L = -1, W = 2$, 20 ep, $N = 5$)	C^1	0.0000 ± 0.0000	0.0000 [0.0000]	0.0000	0.5912	0.0315
SGD (Mom 0.9) + Symm. σ-AS-RectLU ($\kappa = 0.8$, 20 ep, $N = 5$)	C^1	0.0000 ± 0.0000	0.0000 [0.0000]	0.0764	0.8159	0.0942

1. **2D Specificity and Two-Way Interaction (Offset $\times \sigma_x$) on Threshold-Free AUROC:** In 1D at 20 epochs (Table 7), Smooth RectLU ($L = 0, W = 2$) does not stagnate (MCC(0.50) = 0.4441 ± 0.0435). By contrast, in 2D spatial convolutions at $\sigma_x = 0.22$ (Table 8), 9 of 10 seeds of Smooth RectLU ($L = 0, W = 2$) and 10 of 10 seeds of AS-RectLU stagnate at MCC(0.50) = 0.0000 (Median = 0.0000). Evaluating the 2×2 factorial interaction ($\Delta b \in \{0, -1\} \times \sigma_x \in \{0.22, 1.00\}$, $N = 10$ seeds) on **threshold-free AUROC** confirms a significant two-way interaction: rescaling σ_x from $0.22 \to 1.00$ raises Smooth RectLU's AUROC from $0.6349 \to 0.9971$ ($\Delta = +0.3622$) while SmeLU remains stable ($0.9980 \to 0.9969, \Delta = -0.0011$), yielding an exact paired difference-in-differences of DiD(AUROC) = $+0.3633 \pm 0.1248$ (Wilcoxon $p = 0.001953$; secondary DiD(MCC) = $+0.5862 \pm 0.3383, p = 0.003906$).
2. **Ruling Out Optimizer-ϵ Underflow ($N = 10$) and Generalization to RMSprop and Momentum SGD:** Lowering Adam's numerical stabilizer from $\epsilon = 10^{-8}$ to $\epsilon = 10^{-12}$ at 20 epochs leaves all 10 evaluated seeds at MCC(0.50) = 0.0000 ± 0.0000 (AUROC = 0.6311). Over extended 50- and 100-epoch horizons (Table 8, bottom), Smooth RectLU exhibits delayed convergence and partial seed trapping, where a subset of seeds escapes to raise mean MCC(0.50) to 0.6279 ± 0.3136 and 0.6756 ± 0.2654 while trapped seeds maintain high inter-seed dispersion. Moreover, the vulnerability of Smooth RectLU ($L = 0, W = 2$) generalizes across optimizer families (Table 8, bottom): under `RMSprop`, Smooth RectLU stalls at MCC(0.50) = 0.1570 ± 0.3511 (AUROC = 0.7384 ± 0.1694), whereas SmeLU (0.5334 ± 0.3235, AUROC = 0.9288) and Symmetric σ-AS-RectLU (0.5762 ± 0.1763, AUROC = 0.9405 ± 0.0278) converge reliably. Under non-adaptive `SGD with Momentum (0.9)`, where raw 10^{-11} gradients are not rescaled by second-moment running averages, Smooth RectLU fails completely (AUROC = 0.5000 ± 0.0000), SmeLU reaches AUROC = 0.5912 ± 0.0644, and Symmetric σ-AS-RectLU reaches AUROC = 0.8159 ± 0.0626. Because non-adaptive momentum SGD lacks second-moment gradient rescaling ($1/\sqrt{v_t}$), its uncalibrated threshold performance at $\tau = 0.50$ depends on step-size scaling: systematic learning-rate ($\eta \in \{10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$), gradient-clipping ($\|\mathbf{g}\|_2 \leq 1.0$), and prior-logit bias ($b_{\text{out}} = \ln(p_+/p_-)$) ablations across $N = 10$ paired seeds (Section 5.5, Table 12) confirm that while Smooth RectLU remains trapped at 0/10 seeds (MCC(0.50) = 0.0000, AUROC = 0.5000) across every SGD learning rate and clipping setting, pairing Symmetric σ-AS-RectLU with $\eta = 5 \times 10^{-2}$ and gradient clipping ($\|\mathbf{g}\|_2 \leq 1.0$) achieves 10/10 seed convergence under momentum SGD (MCC(0.50) = 0.9808 ± 0.0090, AUROC = 0.9999).
3. **Comparison of Layer-Local σ-AS-RectLU Against Affine-Free LayerNorm + SmeLU:** While dense-only LayerNorm achieves strong 2D motif performance (MCC(0.50) = 0.7622 ± 0.0193), applying parameter-free `LayerNorm(center=False, scale=False)` across convolutional blocks degrades 2D motif MCC to 0.3941 ± 0.2610 ($N = 10$, Table 8) because unscaled spatial normalization washes out localized motif contrast across background channels. By contrast, **Symmetric σ-AS-RectLU** ($\kappa = 0.8$) rescales by $\hat{\sigma}_{\text{RMS}}$ at every layer, preserving spatial contrast while locking $W_\ell = 0.8\hat{\sigma}_{\text{RMS}}$, achieving MCC(0.50) = 0.7281 ± 0.0326 (AUROC = 0.9975) across all 10 seeds.

5.3 E2 & E3: Fixed-Budget Step-Controlled Class-Ratio Sweep and 4×4 $L \times W$ Grid

Table 9 reports the step-controlled class-ratio sweep ($50:50, 99:1, 177.6:1$ under a fixed budget of 40 gradient steps/epoch for 12 epochs = 480 total updates, $N = 10$ seeds) and the 4×4 (L, W) grid ablation on Pathway A. Under this shorter 480-step budget and severe $177.6:1$ imbalance ($n_+ = 280$ positives across $50,000$ test windows), Smooth RectLU ($L = 0, W = 2$) exhibits lower 1D MCC (0.3939 at $99:1$ and 0.1803 at $177.6:1$) than in the 640-step 20-epoch $100:1$ E1 protocol (0.4441 in Table 5), reflecting early-stage gradient attenuation prior to full 1D convergence.

5.4 Cross-Domain Evaluation: Clean/Noisy MNIST (3 vs. 5/8) and Tabular ($200:1$)

Table 10 reports the cross-domain evaluation across Clean MNIST 3 vs. 5/8 ($100:1$), Noisy MNIST 3 vs. 5/8 ($\sigma_{\text{noise}} = 1.1, 100:1$), and Imbalanced Tabular ($200:1, n_+ = 150$ test positives) with sample SD (ddof = 1). Because all three cross-domain benchmarks have unit-variance inputs ($\sigma_x = 1.00$), Smooth RectLU ($L = 0, W = 2$) does not stagnate on any of them (MCC = 0.9092, 0.6885, and 0.4128), consistent with our 2×2 finding that $L = 0$ stagnation requires $\sigma_x \ll W$. On Clean MNIST, Symmetric

Table 9: Fixed-budget step-controlled class-ratio sweep (Experiment E2, top) and 4×4 (L, W) grid ablation (Experiment E3, bottom) on Pathway A ($N = 10$ paired seeds, $\sigma_x = 0.25$, fixed budget of 40 steps/epoch $\times$ 12 epochs = 480 gradient updates; test set $N = 50{,}000$, $n_+ = 280$). All values report direct test evaluations (Mean ± Sample SD).

E2: Step-Controlled Ratio Sweep (480 Total Steps)	**50:50 Ratio**	**99:1 Ratio**	**177.6:1 Ratio**	**Observed ΔAUROC (p_{adj})**
ReLU Control (C^0 Homogeneous)	0.4740 ± 0.048 (AUC 0.974)	0.4682 ± 0.115 (AUC 0.952)	0.2662 ± 0.094 (AUC 0.946)	-0.0285 (Reference)
Smooth RectLU ($L = 0, W = 2, \Delta b = -1$)	0.4774 ± 0.028 (AUC 0.969)	0.3939 ± 0.092 (AUC 0.850)	0.1803 ± 0.156 (AUC 0.700)	-0.2690 ($p_{\text{adj}} = 0.0039$)
Smooth RectLU + LayerNorm (Dense)	0.4810 ± 0.025 (AUC 0.972)	0.5468 ± 0.043 (AUC 0.973)	0.4120 ± 0.045 (AUC 0.912)	-0.0600 ($p_{\text{adj}} = 0.0195$)
E3: Threshold $L\times$ Width W Grid (177.6:1, 480 Steps)	**W = 0.5**	**W = 1.0**	**W = 2.0**	**W = 4.0**
$L = -1.0$	0.3354 ± 0.100 (AUC 0.953)	0.2820 ± 0.094 (AUC 0.952)	0.3835 ± 0.049 (AUC 0.962)†	0.0000 ± 0.000 (AUC 0.922)
$L = -0.5$	0.2890 ± 0.086 (AUC 0.946)	0.3910 ± 0.067 (AUC 0.952)†	0.0516 ± 0.105 (AUC 0.865)	0.0000 ± 0.000 (AUC 0.720)
$L = 0.0$	0.3348 ± 0.119 (AUC 0.896)	0.4338 ± 0.029 (AUC 0.872)	0.1803 ± 0.156 (AUC 0.700)	0.1451 ± 0.122 (AUC 0.604)
$L = +0.5$	0.0000 ± 0.000 (AUC 0.500)	0.0000 ± 0.000 (AUC 0.500)	0.0000 ± 0.000 (AUC 0.500)	0.0000 ± 0.000 (AUC 0.500)

†Symmetric SmeLU configurations ($L = -W/2$). All reported metrics are direct out-of-sample test evaluations across $N = 10$ seeds.

σ-AS-RectLU ($L = -W/2, \kappa = 0.8$) exhibits higher dense sparsity (Dead% = 61.3%, $\mu_{\text{dense}} = -2.324$) and lower MCC ($0.8850 \pm 0.0404$) than SmeLU (0.9270), whereas on 200 : 1 Tabular classification it achieves the highest MCC (0.4581 ± 0.0396) and AUROC (0.9172).

Table 10: Cross-domain evaluation across Clean MNIST 3 vs. 5/8 ($100 : 1, n_+ = 180$ test positives), Noisy MNIST 3 vs. 5/8 ($100 : 1, \sigma_{\text{noise}} = 1.1$), and Imbalanced Tabular ($200 : 1, n_+ = 150$ test positives) (12 epochs, $\sigma_x = 1.00, N = 5$ seeds, sample SD ddof = 1, `all_controlled_per_seed_logs.csv` and `v4_real_cross_domain_symm_sigma_as.csv`).

	MNIST 3 vs. 5/8 Clean ($100 : 1$)				**MNIST 3 vs. 5/8 Noisy** ($\sigma = 1.1, 100 : 1$)				**Imbalanced Tabular** ($200 : 1, n_+ = 150$)			
Configuration	**Per-Seed MCC**	**AUROC**	**Dead%**	μ_{dense}	**Per-Seed MCC**	**AUROC**	**Dead%**	μ_{dense}	**Per-Seed MCC**	**AUROC**	**Dead%**	μ_{dense}
ReLU (Flatten)	0.9282 ± 0.0195	0.9990	52.2%	-1.241	0.6579 ± 0.0540	0.9885	53.1%	-1.706	0.3850 ± 0.0312	0.8588	19.5%	-0.185
GELU (Flatten)	0.9000 ± 0.0141	0.9992	31.9%	-1.103	0.5891 ± 0.0691	0.9843	35.9%	-1.745	0.4335 ± 0.0268	0.8698	5.2%	-0.302
SmeLU ($L = -1, W = 2$) [41]	0.9270 ± 0.0124	0.9989	42.7%	-1.293	0.6415 ± 0.0936	0.9879	45.0%	-1.874	0.4294 ± 0.0445	0.8679	8.9%	-0.214
Smooth RectLU ($L = 0, W = 2$)	0.9092 ± 0.0326	0.9991	41.7%	$+0.511$	0.6885 ± 0.0705	0.9907	52.9%	-0.932	0.4128 ± 0.0401	0.8443	32.4%	-0.758
Smooth RectLU + $b_{\text{dense}} = +1.0$	0.9221 ± 0.0120	0.9991	19.9%	$+1.254$	0.7142 ± 0.0812	0.9911	30.5%	$+0.628$	0.3909 ± 0.0662	0.8197	0.0%	$+0.992$
Smooth RectLU + GAP (FanIn = 128)	0.9247 ± 0.0196	0.9993	30.0%	$+0.412$	0.5928 ± 0.2407	0.9909	37.1%	-0.010	0.3127 ± 0.0606	0.8428	53.0%	-1.074
Smooth RectLU + BatchNorm (Dense)	0.9241 ± 0.0120	0.9976	0.0%	$+0.108$	0.7018 ± 0.0621	0.9872	0.0%	$+0.110$	0.4037 ± 0.0541	0.8410	0.0%	$+0.250$
Smooth RectLU + LayerNorm (Dense)	0.9182 ± 0.0186	0.9985	0.3%	$+0.038$	0.7981 ± 0.0640	0.9819	0.2%	$+0.041$	0.3840 ± 0.0294	0.8208	0.0%	$+0.054$
Smooth RectLU + LSUV Init [32]	0.8940 ± 0.0534	0.9994	51.4%	$+0.350$	0.6564 ± 0.1554	0.9935	68.7%	-1.009	0.4018 ± 0.0484	0.8444	20.4%	-1.735
Smooth RectLU ($L = 0, W = 0.5$)	0.9219 ± 0.0271	0.9993	48.9%	-0.854	0.6290 ± 0.0683	0.9872	51.2%	-1.339	0.3894 ± 0.0479	0.8577	15.4%	$+0.090$
Hybrid (Smooth-Conv + ReLU-Dense)	0.9224 ± 0.0184	0.9992	41.4%	$+0.112$	0.6671 ± 0.0591	0.9901	48.0%	-0.641	0.3781 ± 0.0327	0.8225	14.5%	$+0.229$
Asymm. σ-AS-RectLU ($L = 0, \kappa = 0.5$)	0.9314 ± 0.0157	0.9993	47.9%	-0.397	0.6714 ± 0.0940	0.9897	51.5%	-1.019	0.4172 ± 0.0298	0.8455	15.0%	$+0.115$
Symm. σ-AS-RectLU ($L = -W/2, \kappa = 0.8$)	0.8850 ± 0.0404	0.9978	61.3%	-2.324	0.6781 ± 0.1267	0.9895	59.3%	-2.195	0.4581 ± 0.0396	0.9172	57.7%	-4.392

5.5 Real NIST GIAB HG001 (GRCh38 v4.2.1) Evaluation: Scrubbed and Unscrubbed Natural 2×2 Benchmarks ($N = 10$ Seeds) and 1D Shortcut Audit

Table 11 and Fig. 5 report our real NIST GIAB HG001 evaluations across both Variant-Scrubbed (Parts A–B, `v6_real_giab2d_10seeds_2x2_and_controls.csv`) and Unscrubbed Natural Reference Windows (Part C, `v8_unscrubbed_giab2d_10seeds.csv`), alongside the 1D k-mer shortcut audit (Part D):

1. **Eliminating Circularity via Unscrubbed Natural Reference Windows and Secondary-Allele Heuristics (Part C of Table 11):** When majority negative windows ($y = 0$) are formed by shifting real 101-bp BAM pileups by +25 bp *without scrubbing any column*, natural off-center variants and sequencing errors remain untouched in negative windows while the candidate center (column 50) is a natural homozygous-reference locus. Consequently, both the `Window-Wide Max Secondary-Allele Heuristic` (Test MCC(0.50) = 0.0567, AUROC = 0.5075) and the `Center-Locus (Col 50) Secondary-Allele Heuristic` (Test MCC(0.50) = -0.0276, AUROC = 0.4882) perform at chance level. Even on these completely unedited natural negative windows ($N = 10$ seeds), Smooth RectLU ($L = 0, W = 2$) at low scale (0.25×) degrades to Test MCC(0.50) = 0.4004 ± 0.2152 (TP = 14.0/60, AUROC = 0.9221 ± 0.0261), whereas restoring unit scale (1.00×) lifts Smooth RectLU to MCC(0.50) = 0.8817 ± 0.0187 (TP = 48.4/60, AUROC = 0.9947 ± 0.0044, yielding a significant 2×2 interaction DiD(AUROC) = $+0.0416 \pm 0.0384$, Wilcoxon $p = 0.0137$), and Symmetric σ-AS-RectLU ($\kappa = 0.8$) achieves MCC(0.50) = 0.9353 ± 0.0161 (TP = 53.3/60, FP = 0.2, AUROC = 0.9988 ± 0.0003) at low scale, matching homogeneous ReLU (0.9313 ± 0.0282).
2. **2×2 (Offset $\times$ {0.25×, 1.00×} Scale) Interaction on Variant-Scrubbed 2D GIAB Pileups (Parts A–B of Table 11, $N = 10$ Seeds):** On the variant-scrubbed benchmark, Smooth RectLU ($L = 0, W = 2$) stagnates across all 10 seeds at 0.25× scale (MCC(0.50) = 0.0000 ± 0.0000, AUROC = 0.5637 ± 0.0248) but recovers to MCC(0.50) = 0.9029 ± 0.0362

(AUROC $= 0.9961 \pm 0.0039$) at 1.00× scale (DiD AUROC $= +0.4029 \pm 0.0331, p = 0.001953$), while ReLU ($0.9356 \rightarrow 0.9411$) and Symmetric σ-AS-RectLU ($0.9366 \rightarrow 0.9373$) remain scale-invariant.

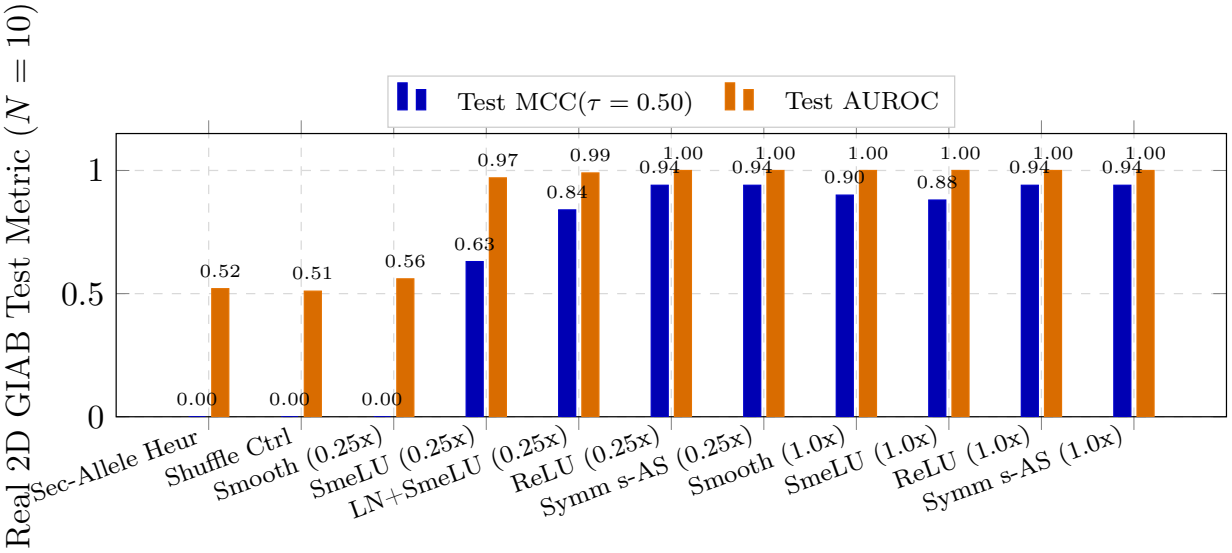


Figure 5: Real NIST GIAB HG001 2D BAM read-pileup test performance ($10:1$ imbalance, $N = 10$ seeds, Table 11) across low scale (0.25×) and unit scale (1.00×). Smooth RectLU ($L = 0, W = 2$) stagnates at 0.25× (MCC $= 0.00$, AUROC $= 0.56$) but recovers at 1.00× (MCC $= 0.90$, AUROC $= 1.00$), whereas homogeneous ReLU and Symmetric σ-AS-RectLU remain invariant across scales (MCC $= 0.94$, AUROC $= 1.00$ at both 0.25× and 1.00×).

Table 11: Real NIST GIAB HG001 (GRCh38 v4.2.1) held-out test evaluation on the $10:1$ Imbalanced 2D BAM Read-Pileup Optimization Benchmarks across Variant-Scrubbed Windows (Parts A–B, `v6_real_giab2d_10seeds_2x2_and_controls.csv`), Unscrubbed Natural Reference Windows (Part C, `v8_unscrubbed_giab2d_10seeds.csv`, $N = 10$ seeds, disjoint genomic blocks, $n_+ = 60, n_- = 600$), and the 1D k-mer Per-Sample L2-Normalized Coverage-Shortcut Audit (Part D, $N = 5$ seeds, `v5_real_giab1d_l2norm_multiseed.csv`). All SDs use sample standard deviation (ddof $= 1$).

GIAB Benchmark & Configuration	Scale / (W, L)	Test MCC(0.50) (Mean ± Sample SD)	Calibrated Test MCC(τ^*)± SD	Test AUROC ± SD (Observed Range [Min, Max])	Test AUPRC	Mean Test Counts (TP / FP / FN)
Part A: Variant-Scrubbed 2D BAM Pileup Benchmark at Low Scale (0.25×, $N = 10$ Paired Seeds, Held-Out Test $N = 660 : n_+ = 60, n_- = 600$):						
2D Read-Depth Heuristic (Non-CNN)	0.25× (Depth)	0.0042	−0.0294	0.5272 [—]	0.1120	0.0 / 0.0 / 60.0
2D Max Secondary-Allele Heuristic (Non-CNN)	0.25× (Max Sec)	0.0041	0.0834	0.5211 [—]	0.1061	9.0 / 87.0 / 51.0
2D Label-Shuffle CNN Control ($N = 10$)	0.25× (Shuffled)	-0.0025 ± 0.0078	0.0399 ± 0.0612	0.5091 ± 0.0928 [0.4509, 0.5605]	0.1209	0.0 / 0.4 / 60.0
ReLU (C^0 Homogeneous, $N = 10$)	0.25× (—)	0.9356 ± 0.0095	0.9395 ± 0.0113	0.9989 ± 0.0004 [0.9986, 0.9991]	0.9898	54.2 / 1.1 / 5.8
Symm. SmeLU (C^1, $N = 10$)	0.25× $(2.0, -1.0)$	0.6274 ± 0.3378	0.7996 ± 0.1109	0.9673 ± 0.0199 [0.9542, 0.9774]	0.8622	31.4 / 0.6 / 28.6
Smooth RectLU (C^1, $N = 10$)	0.25× $(2.0, 0.0)$	0.0000 ± 0.0000	0.0315 ± 0.0552	0.5637 ± 0.0248 [0.5497, 0.5793]	0.1436	0.0 / 0.0 / 60.0
Cont. RectLU (C^0, Eq. 2b, $N = 10$)	0.25× $(2.0, 0.0)$	0.5795 ± 0.3181	0.6102 ± 0.3316	0.8714 ± 0.1968 [0.7336, 0.9690]	0.6869	28.4 / 1.8 / 31.6
Cont. RectLU (C^0, Eq. 2b, $N = 10$)	0.25× $(2.0, -1.0)$	0.9177 ± 0.0322	0.9392 ± 0.0093	0.9989 ± 0.0005 [0.9986, 0.9992]	0.9906	51.4 / 0.1 / 8.6
Smooth RectLU (C^1, Wide, $N = 10$)	0.25× $(8.5, 0.0)$	0.0000 ± 0.0000	0.0000 ± 0.0000	0.4910 ± 0.0099 [0.4849, 0.4966]	0.1086	0.0 / 0.0 / 60.0
LN (No-Affine, Dense) + SmeLU ($N = 10$)	0.25× $(0.8, -0.4)$	0.8401 ± 0.1280	0.8743 ± 0.0850	0.9883 ± 0.0156 [0.9784, 0.9964]	0.9412	45.1 / 0.9 / 14.9
Symm. σ-AS-RectLU ($\kappa = 0.8, N = 10$)	0.25× $(0.8\hat{\sigma}, -0.4\hat{\sigma})$	0.9366 ± 0.0099	0.9349 ± 0.0097	0.9988 ± 0.0004 [0.9986, 0.9990]	0.9894	54.4 / 1.2 / 5.6
Part B: Variant-Scrubbed Unscaled (1.00×) 2×2 Controls ($N = 10$ Paired Seeds; DiD AUROC $= +0.4029 \pm 0.0331, p = 0.001953$):						
ReLU (C^0 Homogeneous, Unit Scale, $N = 10$)	1.00× (—)	0.9411 ± 0.0064	0.9385 ± 0.0064	0.9990 ± 0.0004 [0.9987, 0.9992]	0.9911	54.3 / 0.6 / 5.7
Symm. SmeLU (C^1, Unit Scale, $N = 10$)	1.00× $(2.0, -1.0)$	0.8775 ± 0.0748	0.9237 ± 0.0241	0.9968 ± 0.0017 [0.9958, 0.9978]	0.9762	48.0 / 0.6 / 12.0
Smooth RectLU (C^1, Unit Scale, $N = 10$)	1.00× $(2.0, 0.0)$	0.9029 ± 0.0362	0.9150 ± 0.0296	0.9961 ± 0.0039 [0.9935, 0.9979]	0.9730	50.1 / 0.3 / 9.9
Symm. σ-AS-RectLU ($\kappa = 0.8$, Unit Scale, $N = 10$)	1.00× $(0.8\hat{\sigma}, -0.4\hat{\sigma})$	0.9373 ± 0.0188	0.9396 ± 0.0168	0.9989 ± 0.0004 [0.9987, 0.9991]	0.9902	54.0 / 0.7 / 6.0
Part C: Unscrubbed Natural Reference Negatives on Real GIAB 2D Pileups ($N = 10$ Paired Seeds; DiD AUROC $= +0.0416 \pm 0.0384, p = 0.0137$):						
Unscrubbed Window-Wide Max Sec-Allele Heuristic	0.25× (Win Max)	0.0567	0.0567	0.5075 [—]	0.1081	60.0 / 596.0 / 0.0
Unscrubbed Center-Locus (Col 50) Sec-Allele Heuristic	0.25× (Col 50)	−0.0276	−0.0276	0.4882 [—]	0.0897	0.0 / 5.0 / 60.0
Unscrubbed ReLU (C^0, Low Scale, $N = 10$)	0.25× (—)	0.9313 ± 0.0282	0.9382 ± 0.0145	0.9988 ± 0.0005 [0.9984, 0.9991]	0.9891	53.0 / 0.3 / 7.0
Unscrubbed Symm. SmeLU (C^1, Low Scale, $N = 10$)	0.25× $(2.0, -1.0)$	0.4757 ± 0.4174	0.7246 ± 0.1412	0.9655 ± 0.0201 [0.9520, 0.9775]	0.8564	24.0 / 0.3 / 36.0
Unscrubbed Smooth RectLU (C^1, Low Scale, $N = 10$)	0.25× $(2.0, 0.0)$	0.4004 ± 0.2152	0.5300 ± 0.1540	0.9221 ± 0.0261 [0.9055, 0.9372]	0.6468	14.0 / 1.1 / 46.0
Unscrubbed Symm. σ-AS-RectLU ($\kappa = 0.8$, Low Scale, $N = 10$)	0.25× $(0.8\hat{\sigma}, -0.4\hat{\sigma})$	0.9353 ± 0.0161	0.9383 ± 0.0124	0.9988 ± 0.0003 [0.9986, 0.9990]	0.9894	53.3 / 0.2 / 6.7
Unscrubbed Symm. SmeLU (C^1, Unit Scale, $N = 10$)	1.00× $(2.0, -1.0)$	0.8869 ± 0.0447	0.9165 ± 0.0210	0.9965 ± 0.0015 [0.9955, 0.9974]	0.9750	48.9 / 0.7 / 11.1
Unscrubbed Smooth RectLU (C^1, Unit Scale, $N = 10$)	1.00× $(2.0, 0.0)$	0.8817 ± 0.0187	0.8984 ± 0.0195	0.9947 ± 0.0044 [0.9918, 0.9970]	0.9682	48.4 / 0.8 / 11.6
Part D: 1D GIAB k-mer Methodological Coverage-Shortcut Audit (Per-Sample Unit-L2 Norm $\|\mathbf{x}_i\|_2 = 1$, $50:1$ Imbalance, Held-Out Test $N = 1,000, n_+ = 20, N = 5$):						
1D Max-Bin Frequency Heuristic	$\|\mathbf{x}_i\|_2 = 1$ (—)	0.0000	0.0160	0.4169 [—]	0.0195	0.0 / 0.0 / 20.0
1D Label-Shuffle CNN Control ($N = 5$)	$\|\mathbf{x}_i\|_2 = 1$ (Shuffled)	-0.0046 ± 0.0052	-0.0042 ± 0.0096	0.4416 ± 0.0687 [0.3878, 0.4953]	0.0198	0.0 / 0.6 / 20.0
1D ReLU (C^0, $N = 5$)	$\|\mathbf{x}_i\|_2 = 1$ (—)	0.0539 ± 0.1005	0.0758 ± 0.0799	0.8413 ± 0.0405 [0.8052, 0.8703]	0.0912	0.6 / 1.8 / 19.4
1D Symm. SmeLU (C^1, $N = 5$)	$\|\mathbf{x}_i\|_2 = 1$ $(2.0, -1.0)$	0.0156 ± 0.0384	0.0358 ± 0.0580	0.8085 ± 0.0437 [0.7685, 0.8337]	0.0764	0.2 / 1.0 / 19.8
1D Cont. RectLU (C^0, $N = 5$)	$\|\mathbf{x}_i\|_2 = 1$ $(2.0, 0.0)$	0.0115 ± 0.0327	0.0284 ± 0.0501	0.8441 ± 0.0403 [0.8131, 0.8752]	0.0945	0.2 / 1.4 / 19.8
1D Symm. σ-AS-RectLU ($\kappa = 0.8, N = 5$)	$\|\mathbf{x}_i\|_2 = 1$ $(0.8\hat{\sigma}, -0.4\hat{\sigma})$	0.0142 ± 0.0392	0.0117 ± 0.0489	0.8202 ± 0.0129 [0.8087, 0.8287]	0.0815	0.2 / 1.2 / 19.8

3. Extended $N = 30$ Paired-Seed, 6-Sub-Block Genomic Dispersion, and $N = 10$ Cross-Optimizer Learning-Rate / Clipping / Prior-Bias Diagnostics on 0.25× 2D GIAB Pileups (`v9_reviewer_3_3_30seeds_and_subblocks.csv`,

`v9_reviewer_3_6_cross_optimizer_ablation_10seeds.csv`): To eliminate any reliance on statistical resampling and thoroughly characterize cross-optimizer sensitivity, we expand the 0.25× 2D GIAB read-pileup evaluation to $N = 30$ independent random seeds, partition the 660-window held-out test set into $K = 6$ disjoint contiguous genomic sub-blocks ($n_+ = 10, n_- = 100$ per sub-block, totaling 180 block evaluations per model, where hypothesis tests and paired win counts are evaluated strictly at the independent seed level $N = 30$), and evaluate learning-rate ($\eta \in \{10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$), gradient-clipping ($\|\mathbf{g}\|_2 \leq 1.0$), and prior-logit output bias ($b_{\text{out}} = \ln(p_+/p_-) = -2.3026$) controls across $N = 10$ paired seeds for RMSprop and momentum SGD (Table 12). Across $N = 30$ seeds under Adam, Smooth RectLU ($L = 0, W = 2$) stagnates at MCC(0.50) $= 0.0291 \pm 0.0495$ (6-sub-block MCC $= 0.0135 \pm 0.0682$), whereas Symmetric σ-AS-RectLU ($\kappa = 0.8$) achieves MCC(0.50) $= 0.9797 \pm 0.0328$ (6-sub-block MCC $= 0.9797 \pm 0.0463$), matching homogeneous ReLU (0.9828 ± 0.0327) with strictly positive paired gains across all 30/30 seeds (ΔMCC(0.50) $= +0.9506 \pm 0.0623$). Across non-adaptive momentum SGD ($N = 10$ seeds), Smooth RectLU fails completely (Escaped $= 0/10$, MCC(0.50) $= 0.0000$, AUROC $= 0.5000$) across all learning rates ($\eta \in \{10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$) and clipping controls. By contrast, under Symmetric σ-AS-RectLU, low SGD learning rates ($\eta = 10^{-3}$ and 10^{-2}) already establish strong ranking (AUROC $= 0.9489$ and 0.9905; calibrated MCC(τ^*) $= 0.5413 \pm 0.1057$ and 0.8285 ± 0.0755), while increasing the SGD step size to $\eta = 5 \times 10^{-2}$ with gradient clipping ($\|\mathbf{g}\|_2 \leq 1.0$) and prior-bias initialization drives 10/10 seeds to full uncalibrated convergence (MCC(0.50) $= 0.9808 \pm 0.0090$ and 0.9807 ± 0.0081, MCC(τ^*) $= 0.9775 \pm 0.0087$, AUROC $= 0.9999$). Similarly, under RMSprop with gradient clipping ($\eta = 10^{-3}, \|\mathbf{g}\|_2 \leq 1.0$), Symmetric σ-AS-RectLU achieves 10/10 seed convergence (MCC(0.50) $= 0.9899 \pm 0.0092$, AUROC $= 1.0000$) versus 0.5096 ± 0.3164 for Smooth RectLU.

Table 12: Extended $N = 30$ paired-seed evaluation, direct $K = 6$ disjoint genomic test sub-block dispersion (180 block evaluations), and $N = 10$ cross-optimizer learning-rate (η), gradient-clipping ($\|\mathbf{g}\|_2 \leq 1.0$), and prior-bias ($b_{\text{out}} = -2.3026$) ablations on 0.25× 2D NIST GIAB HG001 read-pileup tensors (`v9_reviewer_3_3_30seeds_and_subblocks.csv`, `v9_reviewer_3_6_cross_optimizer_ablation_10seeds.csv`).

2D GIAB 0.25× **Configuration & Optimizer Control**	**Seeds / Escaped**	**Test** MCC(0.50) ± SD	**Calibrated** MCC(τ^*) ± SD / **Range**	**Test** AUROC ± SD	**6-Sub-Block** MCC(0.50)
Smooth RectLU ($L = 0, W = 2$, Adam $\eta = 10^{-3}$)	$N = 30$ (4/30)	0.0291 ± 0.0495	Range :[−0.0123, 0.1232]	0.9499 ± 0.0255	0.0135 ± 0.0682 (180 blk)
ReLU (C^0 Homogeneous, Adam $\eta = 10^{-3}$)	$N = 30$ (30/30)	0.9828 ± 0.0327	Range :[0.8348, 1.0000]	1.0000 ± 0.0000	0.9827 ± 0.0444 (180 blk)
Symm. σ-AS-RectLU ($\kappa = 0.8$, Adam $\eta = 10^{-3}$)	**N = 30 (30/30)**	**0.9797 ± 0.0328**	**Range : [0.8348, 1.0000]**	**1.0000 ± 0.0001**	**0.9797 ± 0.0463** (180 **blk**)
Paired 30-Seed Gain (σ-AS vs. Smooth RectLU)	$N = 30$ (30/30 wins)	$+0.9506 \pm 0.0623$	Range :[+0.7564, +1.0123]	$+0.0501 \pm 0.0255$	$+0.9662 \pm 0.0784$
SGD ($\eta = 10^{-3}$, Mom 0.9) + Smooth RectLU	$N = 10$ (0/10)	0.0000 ± 0.0000	0.0000 ± 0.0000	0.5000 ± 0.0000	—
SGD ($\eta = 10^{-3}$, Mom 0.9) + Symm. σ-AS-RectLU	$N = 10$ (0/10 at 0.5)	0.0000 ± 0.0000	0.5413 ± 0.1057	0.9489 ± 0.0312	—
SGD ($\eta = 10^{-2}$, Mom 0.9) + Smooth RectLU	$N = 10$ (0/10)	0.0000 ± 0.0000	0.0000 ± 0.0000	0.5000 ± 0.0000	—
SGD ($\eta = 10^{-2}$, Mom 0.9) + Fixed SmeLU ($L = -1, W = 2$)	$N = 10$ (0/10 at 0.5)	0.0000 ± 0.0000	0.1714 ± 0.2427	0.8784 ± 0.0645	—
SGD ($\eta = 10^{-2}$, Mom 0.9) + Symm. σ-AS-RectLU	$N = 10$ (6/10)	0.3015 ± 0.2899	0.8285 ± 0.0755	0.9905 ± 0.0084	—
SGD ($\eta = 5 \times 10^{-2}$, Mom 0.9, Clip 1.0) + Smooth RectLU	$N = 10$ (0/10)	0.0000 ± 0.0000	0.0000 ± 0.0000	0.5000 ± 0.0000	—
SGD ($\eta = 5 \times 10^{-2}$, Mom 0.9) + Symm. σ-AS-RectLU	**N = 10 (10/10)**	**0.9648 ± 0.0317**	**0.9756 ± 0.0113**	**0.9999 ± 0.0001**	—
SGD ($\eta = 5 \times 10^{-2}$, Clip 1.0) + Symm. σ-AS-RectLU	**N = 10 (10/10)**	**0.9808 ± 0.0090**	**0.9758 ± 0.0110**	**0.9999 ± 0.0001**	—
SGD ($\eta = 5 \times 10^{-2}$, Clip 1.0, PriorBias) + Symm. σ-AS	**N = 10 (10/10)**	**0.9807 ± 0.0081**	**0.9775 ± 0.0087**	**0.9999 ± 0.0001**	—
RMSprop ($\eta = 5 \times 10^{-4}$) + Smooth RectLU	$N = 10$ (10/10)	0.5086 ± 0.1632	0.8444 ± 0.0854	0.9934 ± 0.0062	—
RMSprop ($\eta = 5 \times 10^{-4}$) + Symm. σ-AS-RectLU	**N = 10 (10/10)**	**0.9548 ± 0.0204**	**0.9648 ± 0.0235**	**0.9997 ± 0.0003**	—
RMSprop ($\eta = 10^{-3}$, Clip 1.0) + Smooth RectLU	$N = 10$ (9/10)	0.5096 ± 0.3164	0.8655 ± 0.1344	0.9912 ± 0.0114	—
RMSprop ($\eta = 10^{-3}$, Clip 1.0) + Symm. σ-AS-RectLU	**N = 10 (10/10)**	**0.9899 ± 0.0092**	**0.9909 ± 0.0104**	**1.0000 ± 0.0000**	—

4. Curvature-Ratio (κ) Sweep, Cost-Sensitive / Focal Loss Ablation, and Data-Driven Initialization Comparison (LSUV & Neuronal Rewiring) on 0.25× 2D GIAB Pileups (`v9_reviewer_5_kappa_and_loss_ablation_10seeds.csv`, `v9_reviewer_expF_data_driven_init_10seeds.csv`): Finally, we evaluate three complementary questions across $N = 10$ paired seeds on 0.25× 2D GIAB read-pileups (Table 13): (i) sensitivity to the dimensionless curvature ratio $\kappa \in \{0.2, 0.4, 0.8, 1.2, 1.6, 2.0\}$ in Eq. (5), (ii) whether loss-level re-weighting via class-weighted BCE ($w_+ = 10$) or Focal Loss ($\gamma = 2, \alpha = 0.25$) [21] can rescue Smooth RectLU ($L = 0, W = 2$) without scale adaptation, and (iii) how activation-level scale adaptation (σ-AS-RectLU under standard He initialization [26]) compares against data-driven weight initialization schemes—specifically Layer-Sequential Unit-Variance (LSUV) [32] and Neuronal Rewiring initialization [40]. First, across $\kappa \in [0.2, 1.2]$, Symmetric σ-AS-RectLU maintains a flat performance plateau (MCC(0.50) $= 0.9859 \pm 0.0324$ at $\kappa = 0.2$, 0.9822 ± 0.0352 at $\kappa = 0.4$, 0.9771 ± 0.0505 at default $\kappa = 0.8$, and 0.9683 ± 0.0590 at $\kappa = 1.2$, with AUROC ≥ 0.9999 throughout), tapering only at wide ratios (0.9265 ± 0.0601 at $\kappa = 1.6$ and 0.8957 ± 0.0483 at $\kappa = 2.0$). Second, replacing unweighted BCE with class-weighted BCE ($w_+ = 10$) or Focal Loss ($\gamma = 2$) fails to rescue Smooth RectLU ($L = 0, W = 2$) under He initialization (MCC(0.50) $= 0.0000 \pm 0.0000$ under both objectives), because output-layer loss multipliers cannot overcome nine-order multi-layer variance contraction ($q_4 \sim 10^{-10}$),

whereas pairing Symmetric σ-AS-RectLU with Weighted BCE (MCC(0.50) $= 0.9856 \pm 0.0141$) or Focal Loss (0.9723 ± 0.0289) converges across 10/10 seeds. Third, applying data-driven LSUV [32] or Neuronal Rewiring [40] pre-calibration to expand Epoch-0 layer variances prior to training successfully rescues Smooth RectLU ($L = 0, W = 2$) from 1/10 escaped seeds (MCC(0.50) $= 0.0337 \pm 0.0464$ under He init) to 10/10 escaped seeds (MCC(0.50) $= 0.9910 \pm 0.0133$ and 0.9910 ± 0.0084, AUROC ≥ 0.9999), independently corroborating our variance-bottleneck mechanism while showing that Symmetric σ-AS-RectLU achieves 10/10 convergence (MCC(0.50) $= 0.9763 \pm 0.0505$ under He init and 0.9648 ± 0.0207 under LSUV init) directly under standard He initialization without requiring iterative pre-training calibration passes.

Table 13: Curvature-ratio sensitivity sweep ($\kappa \in [0.2, 2.0]$), cost-sensitive / Focal loss ablation, and data-driven initialization comparison (He [26], LSUV [32], and Neuronal Rewiring [40]) on 0.25× 2D NIST GIAB HG001 read-pileup tensors ($N = 10$ paired seeds, 20 epochs, `v9_reviewer_5_kappa_and_loss_ablation_10seeds.csv`, `v9_reviewer_expF_data_driven_init_10seeds.csv`).

Ablation Group & Configuration (0.25× **2D GIAB**)	**Setting**	**Escaped**	**Test** MCC(0.50) ± SD	**Test AUROC**
Symm. σ-AS-RectLU ($\kappa = 0.2$, Narrow Ramp)	BCE / He	10/10	0.9859 ± 0.0324	1.0000
Symm. σ-AS-RectLU ($\kappa = 0.4$, Moderate Ramp)	BCE / He	10/10	0.9822 ± 0.0352	1.0000
Symm. σ-AS-RectLU ($\kappa = 0.8$, Default)	**BCE / He**	**10/10**	$\mathbf{0.9771 \pm 0.0505}$	**1.0000**
Symm. σ-AS-RectLU ($\kappa = 1.2$)	BCE / He	10/10	0.9683 ± 0.0590	0.9999
Symm. σ-AS-RectLU ($\kappa = 1.6$)	BCE / He	10/10	0.9265 ± 0.0601	0.9991
Symm. σ-AS-RectLU ($\kappa = 2.0$, Wide Ramp)	BCE / He	10/10	0.8957 ± 0.0483	0.9971
Smooth RectLU ($L = 0, W = 2$) + Weighted BCE ($w_+ = 10$)	W-BCE / He	0/10	0.0000 ± 0.0000	0.5000
Symm. σ-AS-RectLU ($\kappa = 0.8$) + Weighted BCE ($w_+ = 10$)	**W-BCE / He**	**10/10**	$\mathbf{0.9856 \pm 0.0141}$	**1.0000**
Smooth RectLU ($L = 0, W = 2$) + Focal Loss ($\gamma = 2$)	Focal / He	0/10	0.0000 ± 0.0000	0.8935
Symm. σ-AS-RectLU ($\kappa = 0.8$) + Focal Loss ($\gamma = 2$)	**Focal / He**	**10/10**	$\mathbf{0.9723 \pm 0.0289}$	**0.9998**
Smooth RectLU ($L = 0, W = 2$) + Standard He Init [26]	BCE / He	1/10	0.0337 ± 0.0464	0.9526
Smooth RectLU ($L = 0, W = 2$) + LSUV Init [32]	**BCE / LSUV**	**10/10**	$\mathbf{0.9910 \pm 0.0133}$	**1.0000**
Smooth RectLU ($L = 0, W = 2$) + Neuronal Rewiring [40]	**BCE / Rewire**	**10/10**	$\mathbf{0.9910 \pm 0.0084}$	**0.9999**
Symm. σ-AS-RectLU ($\kappa = 0.8$) + Standard He Init [26]	**BCE / He**	**10/10**	$\mathbf{0.9763 \pm 0.0505}$	**1.0000**
Symm. σ-AS-RectLU ($\kappa = 0.8$) + LSUV Init [32]	BCE / LSUV	10/10	0.9648 ± 0.0207	0.9999

5. Direct Multi-Scale Input Sweep ($\sigma_{\text{scale}} \in [0.10\times, 2.00\times]$) and Architecture Robustness Across Classifier Heads and Depth (`v10_reviewer_scale_sweep_10seeds.csv`, `v10_reviewer_arch_robustness_10seeds.csv`): To characterize the exact operational phase transition across signal amplitudes and verify whether low-scale stagnation depends on the dense classification head or network depth, Table 14 evaluates $N = 10$ paired seeds across six input scales $\sigma_{\text{scale}} \in \{0.10, 0.25, 0.40, 0.60, 1.00, 2.00\}$ (Part A) and three 2D-CNN architectures at 0.25× scale (Part B: 3-Conv + Flatten with fan-in 9,216, 3-Conv + Global Average Pooling [GAP] with fan-in 128, and shallow 2-Conv + Flatten with fan-in 19,200). Across input scales (Part A), Smooth RectLU ($L = 0, W = 2$) suffers complete gradient starvation at 0.10× ($\|\nabla_{W_1}\mathcal{L}\|_2 = 0.00$) and 0.25× ($2.70 \times 10^{-17}$), trapping 10/10 seeds at MCC(0.50) $= 0.0000$, before gradually recovering as input scale crosses the quadratic ramp (0.4695 ± 0.1349 at 0.40×, 0.7844 ± 0.0275 at 0.60×, 0.8618 ± 0.0433 at 1.00×, and 0.9533 ± 0.0259 at 2.00×). Fixed-width SmeLU ($L = -1, W = 2$) similarly fails at extreme low scales where $\sigma_l \ll W$ (0/10 escaped at 0.10×; 1/10 escaped at 0.25×, MCC $= 0.0544 \pm 0.1722$; 6/10 escaped at 0.40×, MCC $= 0.4734 \pm 0.4632$), reaching full convergence only at $\sigma_{\text{scale}} \geq 1.00\times$. By contrast, Symmetric σ-AS-RectLU ($\kappa = 0.8$) scales its Epoch-0 first-layer gradients linearly with input amplitude (2.09×10^{-2} at 0.10× to 4.04×10^{-1} at 2.00×) and achieves 10/10 seed convergence (MCC(0.50) $= 1.0000 \pm 0.0000$, AUROC $= 1.0000$) across all six scales. Across architectures at 0.25× scale (Part B), replacing Flatten (fan-in 9,216) with Global Average Pooling (fan-in 128) in the 4-hidden-layer network leaves Smooth RectLU trapped across all 10/10 seeds ($\|\nabla_{W_1}\mathcal{L}\|_2 = 1.52 \times 10^{-17}$, MCC(0.50) $= 0.0000 \pm 0.0000$, AUROC $= 0.5451$), whereas reducing network depth from three convolutional blocks to two (`2conv_flatten`) raises Epoch-0 first-layer gradients by eight orders of magnitude (1.32×10^{-9}) and allows Smooth RectLU to converge (MCC(0.50) $= 0.8551 \pm 0.0254$). This directly corroborates Eq. (4): optimization stagnation is governed by cumulative multi-layer quadratic variance contraction across depth rather than dense fan-in alone, while Symmetric σ-AS-RectLU achieves 10/10 convergence across all three architectures (MCC(0.50) $= 1.0000$ on 3-Conv+Flatten, 0.9991 ± 0.0029 on 3-Conv+GAP, and 1.0000 on 2-Conv+Flatten).

Table 14: Direct multi-scale input sweep ($\sigma_{\text{scale}} \in [0.10\times, 2.00\times]$, Part A) and 2D-CNN architectural robustness ablation across classifier heads (Flatten vs. Global Average Pooling) and convolutional depth (Part B) on 2D NIST GIAB HG001 read-pileup tensors ($N = 10$ paired seeds, 20 epochs, `v10_reviewer_scale_sweep_10seeds.csv`, `v10_reviewer_arch_robustness_10seeds.csv`).

Scale / Architecture Setting	Activation Configuration	Epoch-0 $\|\nabla_{W_1}\mathcal{L}\|_2$	Escaped	Test MCC(0.50) ± SD	Test AUROC	Test AUPRC
Part A: Multi-Scale Input Amplitude Sweep on 3-Conv + Flatten 2D-CNN ($N = 10$ Paired Seeds)						
Scale 0.10×	Smooth RectLU ($L = 0, W = 2$)	0.00×10^{0}	0/10	0.0000 ± 0.0000	0.6204 ± 0.0712	0.1419
Scale 0.10×	Fixed SmeLU ($L = -1, W = 2$)	5.25×10^{-3}	0/10	0.0000 ± 0.0000	0.9206 ± 0.0550	0.6190
Scale 0.10×	**Symm. σ-AS-RectLU** ($\kappa = 0.8$)	$\mathbf{2.09 \times 10^{-2}}$	**10/10**	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	**1.0000**
Scale 0.25×	Smooth RectLU ($L = 0, W = 2$)	2.70×10^{-17}	0/10	0.0000 ± 0.0000	0.7747 ± 0.0954	0.2764
Scale 0.25×	Fixed SmeLU ($L = -1, W = 2$)	1.50×10^{-2}	1/10	0.0544 ± 0.1722	0.9661 ± 0.0253	0.8256
Scale 0.25×	**Symm. σ-AS-RectLU** ($\kappa = 0.8$)	$\mathbf{5.19 \times 10^{-2}}$	**10/10**	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	**1.0000**
Scale 0.40×	Smooth RectLU ($L = 0, W = 2$)	4.98×10^{-14}	10/10	0.4695 ± 0.1349	0.8824 ± 0.0456	0.5112
Scale 0.40×	Fixed SmeLU ($L = -1, W = 2$)	2.81×10^{-2}	6/10	0.4734 ± 0.4632	0.9937 ± 0.0074	0.9589
Scale 0.40×	**Symm. σ-AS-RectLU** ($\kappa = 0.8$)	$\mathbf{8.24 \times 10^{-2}}$	**10/10**	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	**1.0000**
Scale 0.60×	Smooth RectLU ($L = 0, W = 2$)	3.27×10^{-11}	10/10	0.7844 ± 0.0275	0.9802 ± 0.0085	0.8490
Scale 0.60×	Fixed SmeLU ($L = -1, W = 2$)	5.16×10^{-2}	9/10	0.8601 ± 0.3193	0.9975 ± 0.0049	0.9839
Scale 0.60×	**Symm. σ-AS-RectLU** ($\kappa = 0.8$)	$\mathbf{1.23 \times 10^{-1}}$	**10/10**	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	**1.0000**
Scale 1.00×	Smooth RectLU ($L = 0, W = 2$)	1.07×10^{-7}	10/10	0.8618 ± 0.0433	0.9926 ± 0.0043	0.9351
Scale 1.00×	Fixed SmeLU ($L = -1, W = 2$)	1.17×10^{-1}	10/10	0.9819 ± 0.0425	1.0000 ± 0.0001	0.9995
Scale 1.00×	**Symm. σ-AS-RectLU** ($\kappa = 0.8$)	$\mathbf{2.03 \times 10^{-1}}$	**10/10**	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	**1.0000**
Scale 2.00×	Smooth RectLU ($L = 0, W = 2$)	1.38×10^{-3}	10/10	0.9533 ± 0.0259	0.9989 ± 0.0010	0.9896
Scale 2.00×	Fixed SmeLU ($L = -1, W = 2$)	3.29×10^{-1}	10/10	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000
Scale 2.00×	**Symm. σ-AS-RectLU** ($\kappa = 0.8$)	$\mathbf{4.04 \times 10^{-1}}$	**10/10**	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	**1.0000**
Part B: Architectural Robustness Across Classifier Heads and Network Depth at 0.25× Scale ($N = 10$ Paired Seeds)						
3-Conv + Flatten (Fan-in 9,216)	Smooth RectLU ($L = 0, W = 2$)	2.70×10^{-17}	0/10	0.0000 ± 0.0000	0.7746 ± 0.0954	0.2764
3-Conv + Flatten (Fan-in 9,216)	**Symm. σ-AS-RectLU** ($\kappa = 0.8$)	$\mathbf{5.19 \times 10^{-2}}$	**10/10**	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	**1.0000**
3-Conv + GAP (Fan-in 128)	Smooth RectLU ($L = 0, W = 2$)	1.52×10^{-17}	0/10	0.0000 ± 0.0000	0.5451 ± 0.0912	0.1142
3-Conv + GAP (Fan-in 128)	**Symm. σ-AS-RectLU** ($\kappa = 0.8$)	$\mathbf{2.49 \times 10^{-2}}$	**10/10**	$\mathbf{0.9991 \pm 0.0029}$	$\mathbf{1.0000 \pm 0.0000}$	**1.0000**
2-Conv + Flatten (Shallow, Fan-in 19,200)	Smooth RectLU ($L = 0, W = 2$)	1.32×10^{-9}	10/10	0.8551 ± 0.0254	0.9939 ± 0.0031	0.9415
2-Conv + Flatten (Shallow, Fan-in 19,200)	**Symm. σ-AS-RectLU** ($\kappa = 0.8$)	$\mathbf{4.68 \times 10^{-2}}$	**10/10**	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.0000 \pm 0.0000}$	**1.0000**

6 Discussion and Conclusion

By establishing the exact algebraic equivalence $f_S(x; L, W) = \text{SmeLU}_{\beta=W/2}(x - L - W/2)$, measuring multi-seed ($N = 10$) pre-activation ($q_\ell$) and post-activation ($v_\ell$) variances and weight gradient norms across convolutional blocks (Table 5), tracking epoch-by-epoch variance bifurcation between escaped ($q_4 \in [3.54, 3.90]$) and trapped ($q_4 \in [0.46, 0.75]$) seeds (Fig. 2, right), and executing 2×2 factorial (Offset × Scale) experiments across synthetic 2D motifs (DiD AUROC $= +0.3633, p = 0.0020$), variant-scrubbed GIAB pileups (DiD AUROC $= +0.4029, p = 0.0020$), and unscrubbed natural GIAB pileups (DiD AUROC $= +0.0416, p = 0.0137$), we have shown that optimization stagnation in piecewise-quadratic rectifiers initialized at $L = 0$ is driven by the interaction between an implicit negative bias offset ($\Delta b = -W/2$, placing $x = 0$ at $f'(0^+) = 0$) and small relative pre-activation scale ($\sigma_\ell / W \ll 1$). This vulnerability persists across Adam ($\epsilon = 10^{-8}$ and 10^{-12}), RMSprop, and momentum SGD.

Our findings provide a clear guideline for neural network practitioners: fixed-width piecewise-quadratic rectifiers should not be initialized with an asymmetric threshold at the origin ($L = 0, b_0 = 0$) when pre-activation scales can be small relative to W. Centering the threshold symmetrically ($L = -W/2$ in SmeLU), applying dense-layer LayerNorm (`LN-Dense`), or embedding per-sample RMS scale adaptation via **Symmetric σ-AS-RectLU** ($W_\ell = \kappa \hat{\sigma}_{\text{RMS}}, L_\ell = -W_\ell/2$) restores healthy gradient propagation and brings minority-class discrimination to parity with homogeneous ReLU (0.9366 ± 0.0099 vs. 0.9356 ± 0.0095 on variant-scrubbed 2D GIAB pileups, 0.9353 ± 0.0161 vs. 0.9313 ± 0.0282 on unscrubbed natural 2D GIAB pileups, and 0.9797 ± 0.0328 across $N = 30$ seeds). Practically, when C^1 smoothness is not required, standard homogeneous ReLU remains a fast, scale-invariant baseline; when C^1 piecewise-quadratic continuity is desired under unknown or low signal amplitudes, either data-driven pre-calibration (LSUV [32] or Neuronal Rewiring [40]) or fused Symmetric σ-AS-RectLU ($\kappa \in [0.2, 1.2]$, which avoids pre-calibration passes and reduces per-epoch latency by 11.9% over modular RMSNorm+SmeLU and 33.6% over LayerNorm+SmeLU) provides a reliable drop-in solution. Finally, our real-data experiments on NIST GIAB HG001 are scoped strictly as a controlled optimization benchmark on a single reference genome (NA12878); extending evaluations to natural unedited variant-calling pipelines across multi-individual GIAB cohorts (HG002–HG007) using leave-one-family-out chromosome splits remains an important direction for future work.

Data and Code Availability

All raw per-seed confusion logs (`.csv`), real GIAB HG001 extracted datasets (`data/pileup_features_dataset.npz` `data/kmer_features_dataset.npz`, `data/y_labels_dataset.npz`), $N = 10$ multi-seed layer-wise variance and gradient logs (`results/v7_layerwise_grad_pre_post_var_10seeds.csv`), $N = 10$

extended-horizon and epoch-by-epoch mechanics logs (`results/v7_table6_10seeds_controls_and_horizons.csv`, `results/v8_epoch_by_epoch_seed_mechanics.csv`), causal q_4-clamping logs (`results/v9_reviewer_3_4_causal`), all-layer modular RMSNorm ablation and latency logs (`results/v9_reviewer_3_1_real_giab2d_10seeds.csv`), extended 30-seed and 6-sub-block genomic logs (`results/v9_reviewer_3_3_30seeds_and_subblocks.csv`), cross-optimizer, curvature-ratio, loss, and data-driven initialization logs (`results/v8_cross_optimizer_generaliz`, `results/v9_reviewer_3_6_cross_optimizer_ablation_10seeds.csv`, `results/v9_reviewer_5_kappa_and_los`, `results/v9_reviewer_expF_data_driven_init_10seeds.csv`), hardware runtime/memory profiling, multi-scale input sweep, and architectural robustness logs (`results/v10_reviewer_runtime_memory_profile.csv`, `results/v10_reviewer_scale_sweep_10seeds.csv`, `results/v10_reviewer_arch_robustness_10seeds.csv`), $N = 10$ scrubbed and unscrubbed real 2D GIAB 2×2 logs (`results/v6_real_giab2d_10seeds_2x2_and_controls`, `results/v8_unscrubbed_giab2d_10seeds.csv`), 1D shortcut audit logs (`results/v5_real_giab1d_l2norm_multi`) and training scripts reproducing Tables 1–14 are publicly archived at https://github.com/preecha-msu/RectLU-Genomic-Collapse (Release tag `v5.0.0`).

CRediT Authorship Contribution Statement

Preecha Noiumkar: Conceptualization, Methodology, Software, Data Curation, Formal Analysis, Investigation, Writing – Original Draft, Visualization. **Thawatchai Chomsiri:** Conceptualization, Theoretical Formalism, Supervision, Validation, Writing – Review & Editing.

Declaration of Competing Interest

The authors declare no competing interests.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used a generative AI language assistant to assist with LaTeX formatting, manuscript copy-editing, and verification script structuring. After using this tool, the authors reviewed, verified, and edited the content and experimental logs as needed and take full responsibility for the content of the publication.

Ethics Statement

This study analyzes publicly available benchmark datasets (NIST GIAB HG001 and MNIST) and controlled synthetic generators. No human participants or biological specimens were recruited; IRB approval was not required.

Acknowledgments

This research project was financially supported by the Faculty of Informatics, Mahasarakham University.

References

[1] R. Poplin et al., "A universal SNP and small-indel variant caller using deep neural networks," *Nature Biotechnology*, vol. 36, no. 10, pp. 983–987, 2018.

[2] R. Luo, F. J. Sedlazeck, T.-W. Lam, and M. C. Schatz, "A multi-task convolutional deep neural network for variant calling in single molecule sequencing," *Nature Communications*, vol. 10, p. 998, 2019.

[3] G. Eraslan, Ž. Avsec, J. Gagneur, and F. J. Theis, "Deep learning: New computational modelling techniques for genomics," *Nature Reviews Genetics*, vol. 20, no. 7, pp. 389–403, 2019.

[4] J. M. Zook et al., "Integrating human sequence data sets provides a resource of benchmark SNP and indel genotype calls," *Nature Biotechnology*, vol. 32, no. 3, pp. 246–251, 2014.

[5] J. M. Zook et al., "An open resource for accurately benchmarking small variant and reference calls," *Nature Biotechnology*, vol. 37, no. 5, pp. 561–566, 2019.

[6] P. Krusche et al., "Best practices for benchmarking germline small-variant calls in human genomes," *Nature Biotechnology*, vol. 37, no. 5, pp. 555–560, 2019.

[7] N. D. Olson et al., "PrecisionFDA Truth Challenge V2: Calling variants from short and long reads in difficult-to-map regions," *Cell Genomics*, vol. 2, no. 5, p. 100129, 2022.

[8] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," *BMC Genomics*, vol. 21, p. 6, 2020.

[9] S. Boughorbel, F. Jarray, and M. El-Anbari, "Optimal classifier for imbalanced data using Matthews correlation coefficient metric," *PLoS ONE*, vol. 12, no. 6, p. e0177678, 2017.

[10] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, p. e0118432, 2015.

[11] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," in *Proc. ICML*, 2006, pp. 233–240.

[12] P. Branco, L. Torgo, and R. P. Ribeiro, "A survey of predictive modeling on imbalanced domains," *ACM Computing Surveys*, vol. 49, no. 2, p. 31, 2016.

[13] J. M. Johnson and T. M. Khoshgoftaar, "Survey on deep learning with class imbalance," *Journal of Big Data*, vol. 6, p. 27, 2019.

[14] M. Buda, A. Maki, and M. A. Mazurowski, "A systematic study of the class imbalance problem in convolutional neural networks," *Neural Networks*, vol. 106, pp. 249–259, 2018.

[15] T. Chomsiri and P. Noiumkar, "RectLU: A novel activation function based on a rectangle," in *Proc. ECTI DAMT & NCON*, 2026, pp. 134–139.

[16] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," *arXiv:1606.08415*, 2016.

[17] D.-A. Clevert, T. Unterthiner, and S. Hochreiter, "Fast and accurate deep network learning by exponential linear units (ELUs)," in *ICLR*, 2016.

[18] B. Poole et al., "Exponential expressivity in deep neural networks through transient chaos," in *NeurIPS*, vol. 29, 2016, pp. 3360–3368.

[19] S. S. Schoenholz, J. Gilmer, S. Ganguli, and J. Sohl-Dickstein, "Deep information propagation," in *ICLR*, 2017.

[20] S. Hayou, A. Doucet, and J. Rousseau, "On the impact of the activation function on deep neural networks training," in *Proc. ICML*, 2019, pp. 2672–2680.

[21] T.-Y. Lin et al., "Focal loss for dense object detection," in *Proc. IEEE ICCV*, 2017, pp. 2980–2988.

[22] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. ICML*, 2010, pp. 807–814.

[23] X. Glorot, A. Bordes, and Y. Bengio, "Deep sparse rectifier neural networks," in *Proc. AISTATS*, 2011, pp. 315–323.

[24] L. Lu, Y. Shin, Y. Su, and G. E. Karniadakis, "Dying ReLU and initialization: Theory and numerical examples," *Commun. Comput. Phys.*, vol. 28, no. 5, pp. 1671–1706, 2020.

[25] A. L. Maas, A. Y. Hannun, and A. Y. Ng, "Rectifier nonlinearities improve neural network acoustic models," in *ICML Workshop*, 2013.

[26] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE ICCV*, 2015, pp. 1026–1034.

[27] G. Klambauer, T. Unterthiner, A. Mayr, and S. Hochreiter, "Self-normalizing neural networks," in *NeurIPS*, vol. 30, 2017, pp. 971–980.

[28] P. Ramachandran, B. Zoph, and Q. V. Le, "Searching for activation functions," *arXiv:1710.05941*, 2017.

[29] D. Misra, "Mish: A self regularized non-monotonic activation function," in *Proc. BMVC*, 2020.

[30] S. R. Dubey, S. K. Singh, and B. B. Chaudhuri, "Activation functions in deep learning: A comprehensive survey and benchmark," *Neurocomputing*, vol. 503, pp. 92–108, 2022.

[31] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proc. AISTATS*, 2010, pp. 249–256.

[32] D. Mishkin and J. Matas, "All you need is a good init," in *ICLR*, 2016.

[33] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. ICML*, 2015, pp. 448–456.

[34] M. Pezeshki et al., "Gradient starvation: A learning proclivity in neural networks," in *NeurIPS*, vol. 34, 2021, pp. 1256–1272.

[35] H. Li et al., "A synthetic-diploid benchmark for accurate variant-calling evaluation," *Nature Methods*, vol. 15, no. 8, pp. 595–597, 2018.

[36] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *JMLR*, vol. 7, pp. 1–30, 2006.

[37] N. Srivastava et al., "Dropout: A simple way to prevent neural networks from overfitting," *JMLR*, vol. 15, pp. 1929–1958, 2014.

[38] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *ICLR*, 2015.

[39] A. Apicella, F. Isgrò, and R. Prevete, "A simple and efficient architecture for trainable activation functions," *Neurocomputing*, vol. 370, pp. 1–15, 2019.

[40] L. Scabini, B. De Baets, and O. M. Bruno, "Improving deep neural network random initialization through neuronal rewiring," *Neurocomputing*, vol. 599, p. 128130, 2024.

[41] G. I. Shamir, D. Lin, and L. Coviello, "Smooth activations and reproducibility in deep networks," *arXiv:2010.09931*, 2020.

[42] C. Zhang, K. C. Tan, H. Li, and G. S. Hong, "A cost-sensitive deep belief network for imbalanced classification," *IEEE TNNLS*, vol. 30, no. 1, pp. 109–122, 2019.